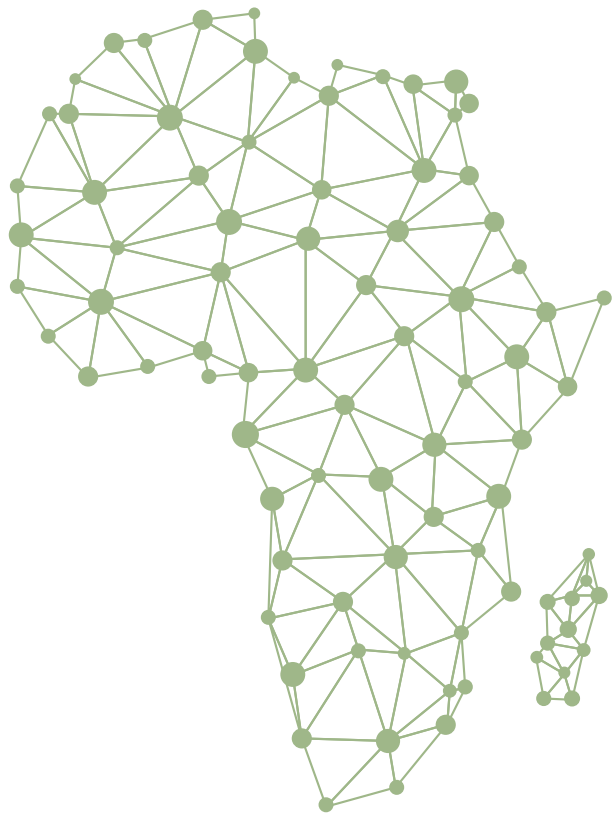


OpenMod Africa

Specification of the Eastern Africa Case Studies

DELIVERABLE NO. 6.1



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List of Acronyms

AfDB	African Development Bank
BAU	Business-as-usual
CRGE	Climate Resilient Green Economy
CTS	Current Trend Scenario
DRC	Democratic Republic of Congo
EAPP	Eastern Africa Power Pool
EB	Electric Bus
ECP	Electric Connections Policy
EEP	Ethiopian Electric Power
EEU	Ethiopian Electric Utility
EV	Electric Vehicle
GDS	Grid Densification Scenario
GDP	Gross Domestic Product
GHG	Greenhouse Gas
ICE	Internal Combustion Engine
ICT	Information and Communication Technologies
INEP	Integrated National Energy Plan
IPP	Independent Power Producer
KETRACO	Kenya Electricity Transmission Company Limited
KNES	Kenya National Electrification Strategy
KP	Kenya Power
LPG	Liquified Petroleum Gas
MTF	Multi-Tier Framework
NDC	Nationally Determined Contributions
NEP 2.0	National Electrification Plan 2.0
OM4A	OpenMod4Africa
PUE	Productive Use of Energy
SAAP	Southern Africa Power Pool
SHS	Solar Home Systems
SME	Small & Medium Enterprises
UEGCL	Uganda Electricity Generation Company Limited
UETCL	Uganda Electricity Transmission Company Limited
VRE	Variable Renewable Energy
WP	Work Package



Executive Summary

The report focuses on defining the specifications for East African case studies, i.e., establishing the methodologies, assumptions, and objectives for case studies in Ethiopia, Kenya, Uganda, and the broader Eastern Africa Power Pool (EAPP) region. These case studies are designed to guide the planning and development of secure, sustainable, resilient, and cost-effective energy systems; they aim to address critical energy challenges at both national and regional levels.

For Ethiopia, the focus is on national and city-level energy system analyses. Kenya and Uganda's case studies replicate Ethiopia's approach while tailoring methodologies to their specific needs, such as enhancing renewable energy integration, expanding electrification, and optimizing infrastructure planning. At the regional level, the EAPP case studies will explore cross-border interconnections, trade opportunities, and harmonized policies to maximize resource utilization and improve energy security across member countries.

The report outlines the framework and tools to be used in these studies, including GENeSYS-MOD, openTEPES, OnSSET, and GIS-based models. It emphasizes the importance of scenario development, with analyses ranging from Business-as-Usual (BAU) scenarios to high renewable energy integration, climate resilience, and efficiency improvement scenarios. Key modelling assumptions include projections for demand growth, cost trajectories for renewable energy technologies, and hydrological variability. Sensitivity analyses will be employed to address uncertainties in these parameters.

By defining detailed specifications, this report lays the groundwork for implementing the case studies and enables the application of the OM4A toolbox in diverse contexts. The results will support regional energy and power systems planning, policy formulation, and investment decisions aimed at achieving universal access to clean, reliable, and affordable energy. Through these efforts, the OM4A project seeks to contribute to the sustainable development of energy systems in Eastern Africa while enabling capacity building and knowledge sharing through the OM4A toolbox.

1. Introduction

The Horizon Europe Energy project 101118123 – OpenMod4Africa (Open Modelling Toolbox for development of long-term pathways for the energy system in Africa) is a project that aims to develop an Open Energy System Modelling Toolbox (OM4A Toolbox) that is adapted to African needs and uses.

One of the key objectives of the OpenMod4Africa project is to provide new insights into the possible pathways for the development of the African energy system. OpenMod4Africa partners from the Western Africa and Eastern Africa Power Pool (EAPP) regions will conduct regional case studies under Work Packages (WP) 5 and 6, respectively, of this project to enable this objective. The case studies are comprised of various energy system and power system analyses at the country and regional levels; they will test and demonstrate the use of the toolbox while also providing new insights on the development of a secure, competitive, and sustainable energy and power supply in the regions. Specifically, the Eastern Africa Case Study (WP6) will include studies at the EAPP, as well as analyses of specific national and local energy systems, including Ethiopia, Kenya, Uganda, and the city of Addis Ababa.

The objectives under WP6 are the following:

1. Build knowledge among academia, researchers, and decision makers in the EAPP region in using the OM4A Toolbox for analysis of the development of the regional energy and power systems.
2. Build knowledge among policy and energy decision makers in the EAPP region about the OM4A Toolbox and how energy system modelling can contribute to developing clean and cost-efficient energy and power systems.
3. Provide new insights about how to develop secure, competitive, and sustainable energy and power supply in the EAPP region. The analyses shall particularly focus on the integration of renewables.

In line with objectives 1 and 3 of WP6, this deliverable develops and outlines the ‘Specifications of the Eastern African Case Studies’. This deliverable also builds on discussions and analyses conducted under Task 2.4, which have identified the main challenges in the EAPP region in close dialogue with relevant stakeholders. The document is composed of three sections. The first section provides an overview of the current energy systems in the geographic focus areas for the Eastern Africa case studies – Ethiopia, Kenya, Uganda, and the broader EAPP region. Additionally, this first section includes a sub-section that details the relevant energy and power system modelling studies conducted so far within this same geography.

The second section of the report outlines key modelling focus areas and relevant characteristics of the EAPP region based on the analysis of the existing energy systems, national targets, and challenges. This is informed by the review of previous works and reports, the characteristics of the region previously identified, as well as a first comparison among the three countries included under WP6. Lastly, the third section of the deliverable provides the specification of the main case studies to be conducted under WP6. For each of the planned case studies under WP6, this section provides a general description of the planned analyses and objectives, as well as the relevant assumptions, scenarios, expected outcomes, and other details. The case studies will include country-level energy system analysis using GENeSYS-MOD to evaluate generation expansion pathways up to 2050; each country will have sub-national geographic granularity at the level of administrative regions. Rural electrification case studies are also planned for the three countries through the OnSSET tool. Additionally, EV-PV will be used to evaluate optimal charging infrastructure deployment for electric mobility use cases across urban cities. Power system analysis will also be conducted at the regional power pool level (i.e., EAPP) to assess optimal expansion scenarios over a 10-year time horizon for the 13 member countries.

2. Overview of Energy Systems

2.1. Eastern Africa Power Pool Region (EAPP)

The EAPP comprises 13 member countries, with 16 utilities, serving approximately 600 million people. The region's population accounts for approximately 40% of Africa's total population. The EAPP region is rich in clean energy resources such as hydropower, geothermal, variable renewable energy (VRE) resources (solar, wind, ...), as well as oil and gas. Despite this large energy potential, the region is characterized by a low electricity access rate, apart from Egypt and Libya, which have achieved universal access. The EAPP region is also characterized by its large population and strong economic growth. The average regional growth rate in 2019 was nearly double the 3.4% average for Africa. Some of the relevant characteristics of the EAPP region's energy systems are provided below:

- **Low electricity access rates** – The average electricity access rate for the region is 48%, with a minimum of 7.7% in South Sudan. Notably, Egypt and Libya have achieved universal access, while Kenya has also achieved a relatively higher access rate of over 80%. Biomass is the predominant source of energy utilized in countries with low access rates.
- **Wide variation in the cost of electricity** – The cost of electricity varies widely across the region, ranging from being below USD 0.06/kWh in Ethiopia and Egypt to being greater than USD 0.17/kWh in Rwanda, and nearly USD 1.00/kWh in Somalia. This variation is due to multiple factors, such as the generation technology mix and the scale of production.
- **Share of renewable penetration** – The scale of renewable generation penetration varies significantly across the EAPP region due to differences in the level of development of grid infrastructure and the availability of energy resources. Hydropower is dominant in countries such as the Democratic Republic of Congo (DRC), Ethiopia, and Uganda, where it makes up 86-100% of the generation. In contrast, in Djibouti, Egypt, Libya, and South Sudan, over 90% of their installed generation is non-renewable. In regions where national grid infrastructure is minimal, there are also cases of widespread diesel mini-grids; Somalia is one such example with a decentralized energy landscape serviced through isolated mini-grids powered by diesel generators.

2.1.1. Generation

The EAPP covers a large geographical area characterised by a diverse power generation mix (e.g., natural gas), which is predominant in the North, hydro in the middle, and this is mixed in the South (i.e., natural gas and hydro).

The amount of installed generation capacity in the region was about 85.6 GW per the 2022 EAPP data. The existing generation capacity mix in EAPP is dominated by natural gas and hydro generation, which contribute 91% to the total capacity mix. Egypt contributes about 84% to the EAPP's installed generation capacity from natural gas and has the largest amount of installed generation capacity (around a 70% share of the installed capacity in EAPP). In contrast, nearly 91% of Ethiopia's installed generation capacity is from hydropower. Kenya, along with Egypt, has the most diverse capacity mix in EAPP. Figure 1 below summarises the total installed generation capacity (MW) in EAPP per country and technology.

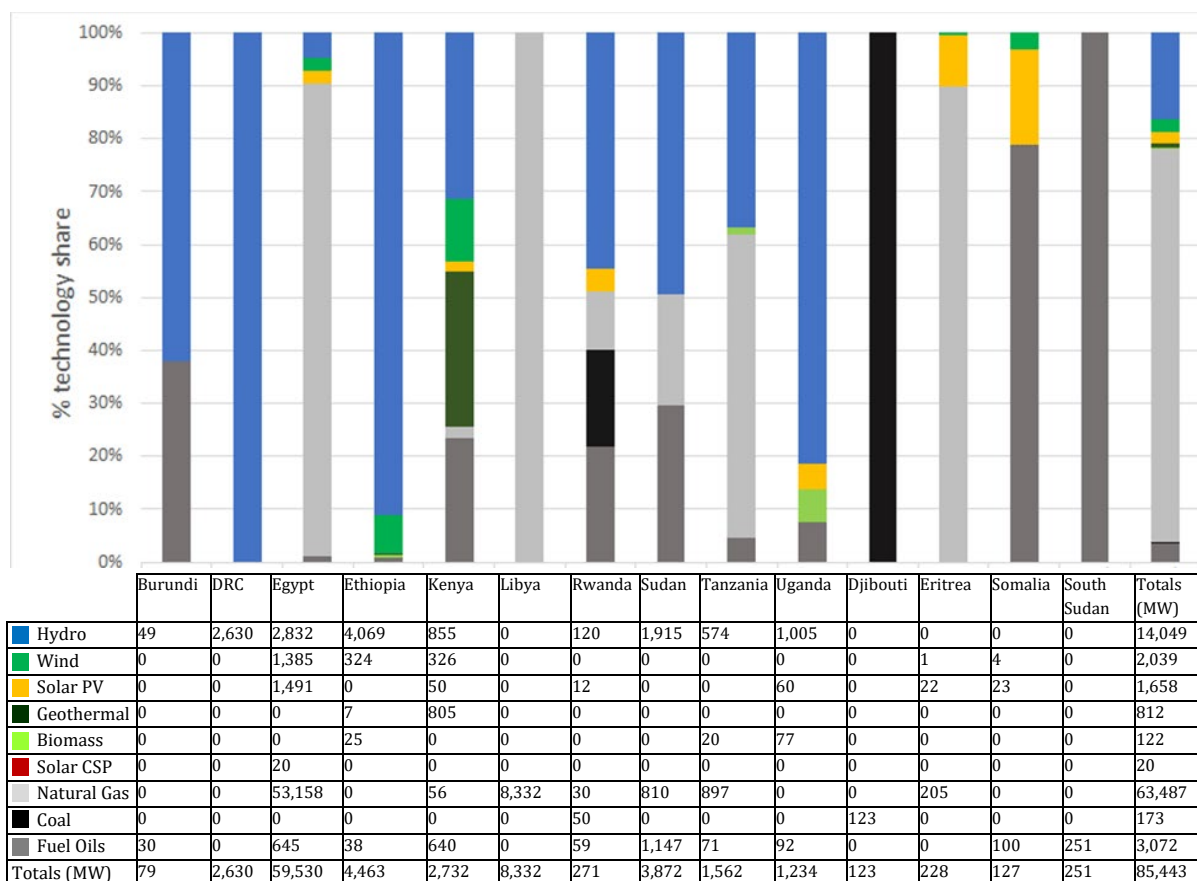


Figure 1: EAPP – Summary of Installed Power Plant Capacity Across the EAPP

Hydro resources contribute about 75% of the overall capacity using renewable energy resources within EAPP. Notably, the DRC and Ethiopia have almost 100% of their generation capacity corresponding to renewable energy resources.

2.1.2. Demand forecast

The electricity demand in EAPP is projected to grow rapidly, driven by the expected economic growth in the region and the plans to increase the electrification rates in the region. In 2022, the total forecasted peak was 59.4 GW, which is expected to grow to 96.8 GW by 2030. This trajectory is aligned with country-level plans and is based on historical growth trends. Figure 2 below demonstrates the projected growth in annual electricity demand and peak demand within the EAPP up to 2030. Population growth, growing urbanization rates, and increased consumption per capita are seen as major drivers for electricity demand growth in the region.

2.1.3. Interconnection

The combination of hydroelectric power plants and fossil fuel-based power stations results in an important spread in marginal generation costs, therefore, creating a strong incentive for cross-border power exchanges (aiming at maximizing the benefits of a diverse generation portfolio – “portfolio effect”). The benefits from cross-border power exchanges tend to increase with the planned expansion of variable renewable energy generation (mainly wind and solar power).

It is expected that the entire EAPP area will be interconnected by 2025. The total capacity of the internal EAPP cross-border lines was ca. 4 GW in 2023; this capacity is expected to increase to 4.8 GW in 2025 and 14.1 GW in 2031. With existing and nearly completed under-construction transmission capacities, the EAPP will be divided into the following three “islands”, also shown below in Figure 3:

1. The DRC (West and South)
2. Burundi, Rwanda, DRC (East), Egypt, Libya, Sudan, Ethiopia, Djibouti, Kenya, Uganda, Tanzania, Ethiopia, Sudan, South Sudan
3. Somalia

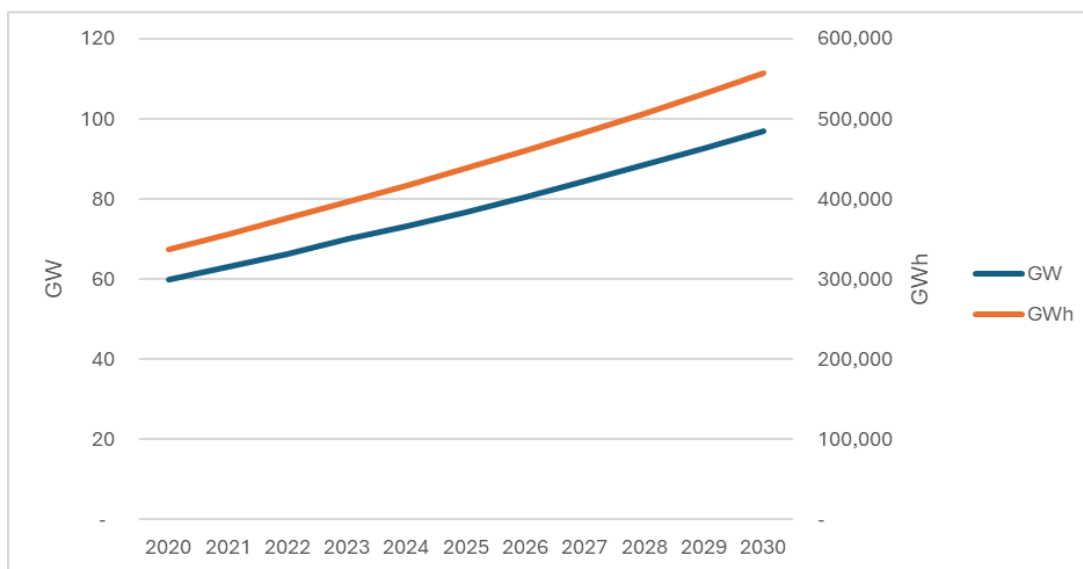


Figure 2: Expected development in the EAPP yearly energy and peak demand

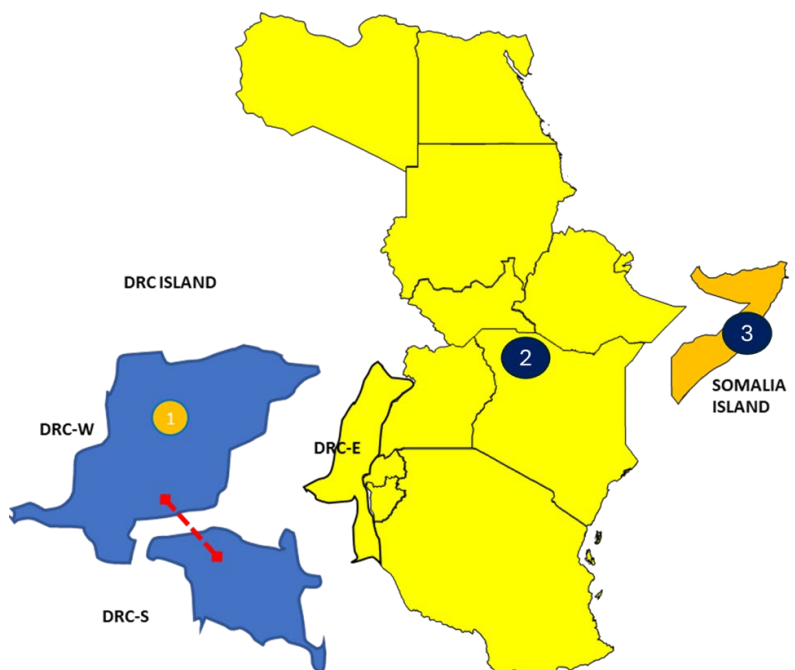


Figure 3: EAPP – Electric islands

The EAPP electric islands in Figure 3 represent existing and nearly completed under-construction cross-border transmission corridors; note that the DRC is divided into both West and South, as well as East.

There will be five synchronous areas, as shown in Figure 4. DRC West and DRC South are two synchronous systems connected through a DC line. In addition, because of the DC-link between Ethiopia and Kenya, two additional synchronous areas will be formed. Kenya, Tanzania, Uganda, Rwanda, DRC-E, and Burundi are in one synchronous system, while Ethiopia, Djibouti, Sudan, South Sudan, Egypt, and Libya are in another synchronous system. The Somalia system is also considered a separate synchronous area.

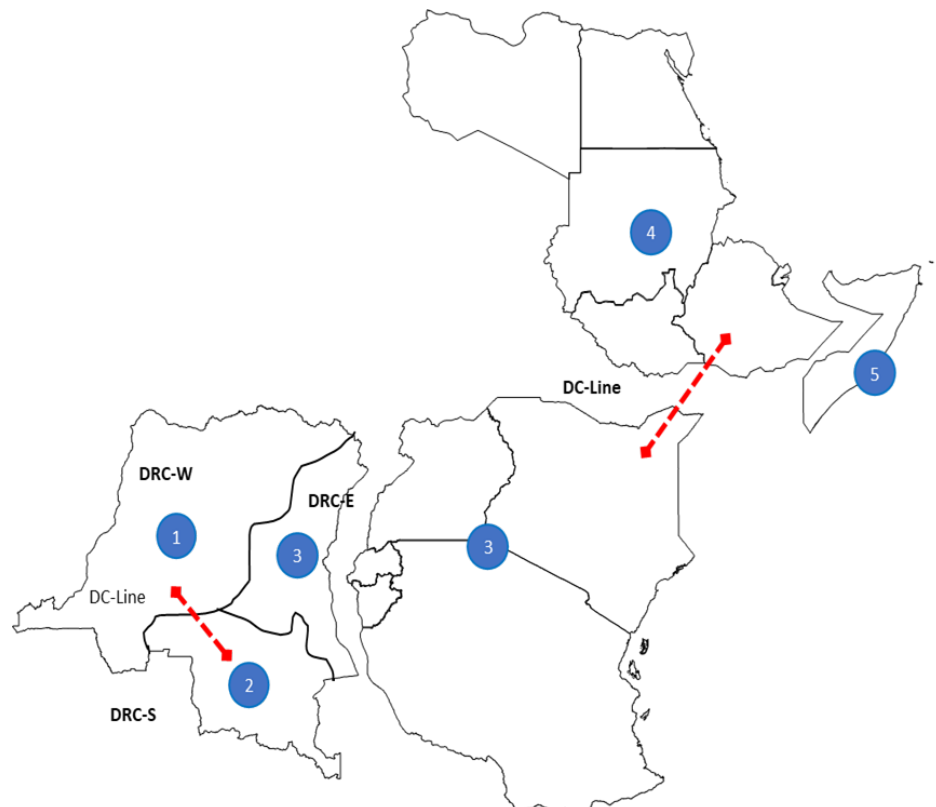


Figure 4: Synchronous areas in the EAPP.

Note that the DRC will be divided into three synchronous areas: West, South, and East. DRC East will be synchronous with the southern synchronous area.

The DRC-west and south are not connected to the rest of the EAPP system at present, and the Somalia power grid is still developing; for this reason, officially, the EAPP considers that there are two active synchronous areas: these are the Northern and Southern, separated by the DC link between Ethiopia and Kenya.

The scenarios and modelling focus areas for the EAPP case study should support the undertaking of sensitivity analyses for critical parameters where data is uncertain. These include variables such as the discount factor, socio-economic growth, urbanization, storage needs, technology learning rate, local economics of renewable technologies, as well as policy targets set at the national and regional levels.

2.2. Ethiopia

Ethiopia has a population of 126.5 million as of 2023 [1]; it ranks second among the most populous nations in Africa after Nigeria. Over the past 15 years, Ethiopia has sustained an average annual economic growth rate of nearly 6% and its economy relies primarily on agriculture, followed by industry, services, mining, and tourism. With a rapidly growing population and expanding economy, Ethiopia is increasingly focusing on enhancing energy access and security to drive sustainable development across various sectors. Ethiopia requires substantial investments to meet its ambitious generation and electrification targets.

Generation: Ethiopia's renewable energy potential is estimated to be more than 60,000 MW. However, the currently exploited levels of renewable energy represent less than 10% for hydropower, and less than 1% for geothermal, wind, and solar resources. Ethiopia's renewable resource potential, outlined in the country's latest National Electrification Plan (NEP 2.0) [2], is provided in Table 1.

Table 1: Energy resource potential for Ethiopia [3]

Resource	Unit	Exploitable Reserve	Exploited Percent
Hydropower	MW	45,000	< 10%
Solar/day	kWh/ 2	Avg. 5.5	< 1%
Wind power	GW	100	< 1%
Wind speed	m/s	> 6.5	
Geothermal	MW	7,000	< 1%
Wood	Million Ons	1,120	50%
Agricultural waste	Million Ons	15-20	30%
Natural gas	Billion m3	113	0%
Coal	Million tons	300	0%
Oil shale	Million tons	253	0%

Ethiopia's installed energy generation capacity, last reported at the end of 2023, was 5,222 MW, with hydropower sources accounting for 90.7% of the country's electricity production. Ethiopia's 10-Year Development Plan aims to increase the total installed generation capacity to 19,900 MW by 2030. A breakdown of the currently installed capacity and the projected installed capacity for 2030 is provided in Figure 5. For 2030, the figure provides details for 18,186 MW of the projected 19,900 MW capacity; this figure does not include new diesel, biomass, or undisclosed renewable energy projects.

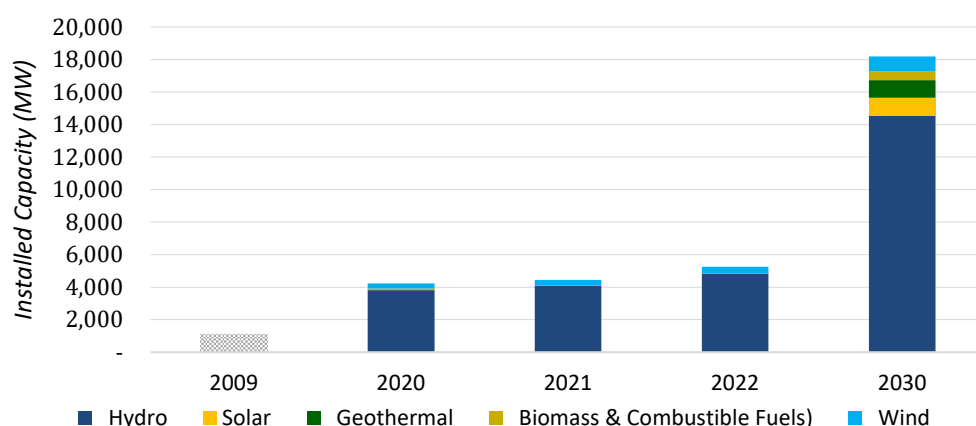


Figure 5: Current and 2030 projected installed generation capacity per technology, Ethiopia [4]

These investments would, if realized, help in broadening the mix of energy sources that Ethiopia would rely on; the diversification of sources beyond hydro is a purposive strategy at the core of the expansion plans. The government projects that Independent Power Producers (IPPs) will develop new geothermal, hydro, solar, and wind power generation projects, with the public sector taking on regulatory and off-taker roles, thus contributing to the achievement of the 2030 targets [3]. However, this planned increase in generation capacity may still not be sufficient to meet electricity demand in 2030 due to the projected increase in population, electrification, and economic development.

Demand/Consumption: Households account for 88% of the total national energy consumption, followed by the transport sector consuming 8.4% (petroleum & ethanol) and industry and construction consuming 2.3% of total energy supply, as shown in Figure 6 below [5]. A large share of this household energy demand is met using traditional biomass resources. Electricity generation supplies only 2.6% of the country's final energy demand (shown in Figure 7).

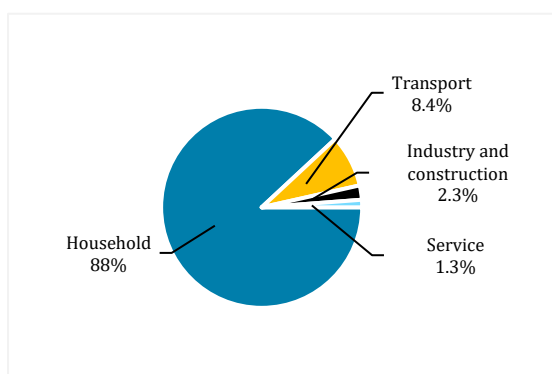


Figure 6: Energy consumption by consumer type in Ethiopia, 2023

Currently, Ethiopia's electricity consumption per capita stands at 100 kWh/year, whereas the average per capita electricity consumption in sub-Saharan Africa is 521 kWh [4]. Ethiopian Electric Utility (EEU), the public utility which provides electricity distribution and retail services, has 4.3 million customers, of which 3.8 million (89%) are household customers. Moreover, households, in aggregate, consume more electricity than industrial and commercial consumers combined (see Figure 8). Notably, only 21% of the energy consumption by the industry and the construction sector corresponds to electricity; the remaining is supplied by petroleum and coal plants.

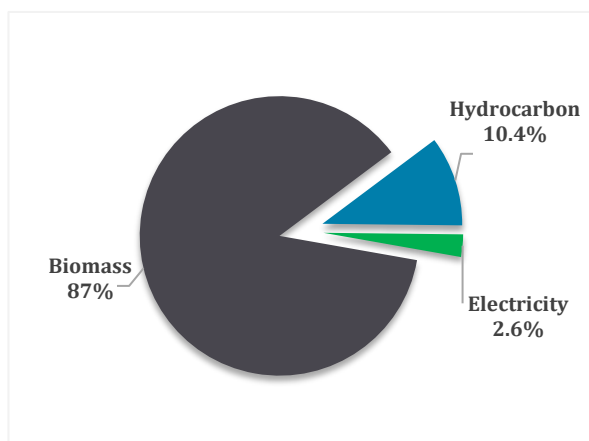


Figure 7: Total energy supply in Ethiopia, 2023 [6]

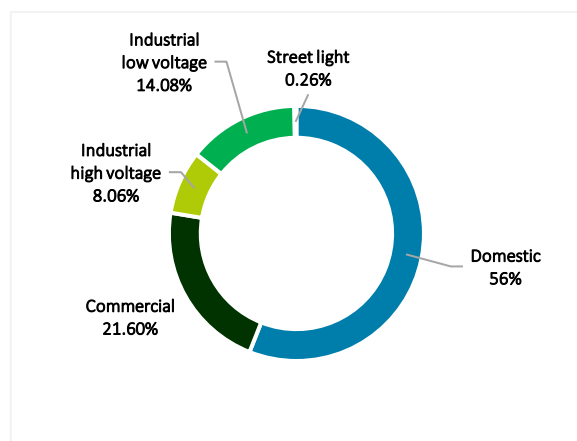


Figure 8: Electricity consumption by tariff category, Ethiopia, 2023 [6]

Transmission & Distribution: With respect to grid infrastructure, Ethiopia's high-voltage transmission lines extend over 20,634 km [5], with a plan to expand the transmission lines to 29,900 km. The voltage

level of the transmission lines ranges from 132kV to 500kV. The distribution lines extend over ~134,000 km, comprising 73,000 km of medium voltage distribution lines (15/33kV), and 60,704 km of low voltage distribution lines (380/220V) [6]. According to NEP 2.0, the distribution network has been extended to reach close to 80% of the population and now covers 60% of the country's geographic areas as of 2018. Nevertheless, the last-mile connection has not kept pace with expansion efforts.

Currently, while 99.9% of Addis Ababa's population has access to electricity, the majority of rural households rely on traditional biomass and other off-grid solutions. Electricity access rates per population category are provided in Table 2.

Table 2: Electricity Access-Population Category [2]

Area	Electricity access rate
Deep rural area (25km> MV line)	5%
Rural Area	5-10%
Peri-urban area	20-30%
Urban Area	96%

Electrification Strategy: The Ethiopian government planned to achieve universal electricity access by 2025 through its NEP 2.0. Through NEP 2.0, EEU leads the grid expansion and strengthening works, whereas the private sector and various ministries were expected to collaborate to achieve the implementation of off-grid access plans. To support the off-grid targets, the utility is also constructing an additional 201 mini-grids across rural locations. While there has been significant progress to increase distribution networks across rural areas, current progress is not on track to meet the 2025 target.

Table 3: NEP 2.0 Objectives [3]

NEP 2.0 Programs	Number of connections (Millions of Households)
Grid Connections	8.2 (new connections) – 65% of entire population to have access to electricity on-grid
Off-Grid Connections	6 (new connections) – 35% of the population to have access to electricity through off-grid solutions
Regularization ¹	3.8

Through ongoing and upcoming programs, EEP and EEU are set to support the grid expansion and strengthening, increasing formal utility connections from 4.5 million in 2022 to 15 million in 2032. This growth aims to achieve universal access by 2035 and 96% grid-based access by 2040 [6].

2.3. Kenya

Kenya has a population of approximately 55 million people as of 2022. With an average annual economic growth rate that ranges between 5% and 6%, its growth has been bolstered by several sectors, including agriculture, tourism, manufacturing, information and communication technology, and other services. Kenya has ambitious development objectives, including its Vision 2030 initiative, and aims to transition into a newly industrialized, middle-income nation by 2030. Key areas of focus encompass economic diversification, infrastructure enhancement, education, healthcare, and governance reforms.

¹ Regularization refers to the new, metered connections added to the grid for customers that were previously connected to the grid and paying for electricity, but without a metered connection and often through unsafe service drops.

Generation: Kenya has significant renewable energy potential, of which only a limited share has been utilized, as shown in Table 4 below.

Table 4: Renewable energy resource potential for Kenya [7]

Resource	Unit	Exploitable Reserve	Exploited Percent
Geothermal	GW	7-10 GW	<10%
Hydropower	GW	3 GW	28%
Solar Irradiation	kWh/m2	4-6 kWh/m2	1%
Wind Speed	m/s	6 - 9.5 m/s	-
Wind Potential [8]	GW	3 GW	15%

Much of the power generation in Kenya is based on renewable sources, with significant contributions from geothermal and hydropower resources. The third highest generation type is from thermal sources (Gas Turbines and Medium Speed Diesel). Kenya has surplus installed capacity to meet its current grid demand, including a peak reserve margin of 30 percent [9]. Based on the 2023 report by Kenya's Electricity Transmission Company (KETRACO), Kenya's generation capacity is primarily supplied by KenGen, its national power producer, which owns nearly 60 percent of the 2,967 MW total effective generation capacity. Additional capacity belongs to Independent Power Producers and the government's rural electrification program, which accounts for nearly 39 percent and 1 percent of the total effective generation capacity, respectively. The breakdown of its on-grid and off-grid generation through the different generation sources is provided in Table 5.

Table 5: Kenya's generation mix as of December 2022

Technology	Installed Capacity (MW)	Effective*/Contracted (MW)
<i>Hydro</i>	839	810
<i>Geothermal</i>	950	871
<i>Thermal (MSD)</i>	586	566
<i>Thermal (GT)</i>	60	56
<i>Wind</i>	436	426
<i>Biomass</i>	2	2
<i>Solar</i>	210	210
<i>Import</i>	200	0
Total - Interconnected System	3,283 MW	2,942 MW
<i>Off-grid thermal</i>	35.6	23.03
<i>Off-grid Solar</i>	2.26	1.92
<i>Off-grid Wind</i>	0.55	0
Total Capacity MW	3,322 MW	2,967 MW

Kenya's annual report shows total electrical energy generation in the 6 months of July to December 2022 was 6,670.9 GWh [10]. Geothermal and hydropower accounted for ~46 percent and ~22 percent of this

total, respectively, while thermal sources accounted for 10 percent. It is interesting to note that imports only accounted for approximately 2 percent of the total energy generated/consumed. KenGen recently announced a renewed plan to scale up the deployment of renewable energy in the country by adding an additional 3,000 MW within the next 10 years, of which 2,000 MW will be from geothermal and hydropower sources. Existing power plants will also be rehabilitated to make them more efficient for sustainable generation.

Demand/Consumption: Nearly 98 percent of all customers on the grid are households. However, they only account for roughly 31 percent of total energy consumption. Industrial customers represent the largest energy consumption category at 52 percent of total energy consumption. Commercial consumers (small and medium enterprises (SMEs)) also represent an increasing demand category, accounting for 16 percent of total energy consumption. Although electric mobility is at a very nascent stage in Kenya, charging demand is estimated to account for 0.01% of consumption.

Transmission & Distribution: KETRACO oversees the construction and upkeep of the country's national power transmission grid. The transmission network comprises 90 substations and has a total length of approximately 9,011km [11]. Of this, around 5,476km (about 60.77%) of the lines and 39 substations with a combined capacity of 5,928 MVA are owned and managed by KETRACO. Kenya faces an approximate 16% loss of generated power due to the aging transmission and distribution networks [12]. By 2039, KETRACO is expected to have deployed around 11,246 km of transmission lines and 129 substations.

Kenya has also worked to aggressively increase access to the grid, having more than doubled electricity access from 32% in 2013 to 75% of households in 2022. 100% of urban areas in Kenya have access to electricity, while access in rural areas stands at 65%. Kenya Power (KP) is the sole distribution company in Kenya. In 2023, KP bought 6,805 GWh and sold 5,226.9 GWh, the difference accounting for technical losses (transmission & distribution), and commercial losses (unauthorized connections, unauthorized electricity consumption, and meter manipulation).

Electrification Strategy: There are several significant initiatives aimed at expanding electricity access across Kenya. In 2018, Kenya developed its National Electrification Strategy (KNES), which aimed to reach universal electricity access by 2022. This was planned through a combination of measures: grid expansion, intensification & densification, mini-grid development, and stand-alone solar home systems. While this 2022 goal of universal electrification has not been achieved, KNES has enabled the country to achieve the highest electrification rate amongst its East African neighbours. The challenges faced during the KNES implementation included high costs of grid extension, grid intensification, and connection charges.

A second initiative aimed at increasing electrification is the Last Mile Connectivity Project - a government initiative aimed at increasing electricity access by expanding distribution lines, particularly in rural areas, and affordably connecting households. The project sought to address the high cost of connection in rural areas through subsidies and loans.

2.4. Uganda

Uganda has a population of approximately 49.64 million people (2024). Over the past decade, the country has maintained an average annual economic growth rate of around 5.3%, driven by investments in sectors such as agriculture, tourism, and energy. Through its recent Energy Transition Plan, Uganda has set a net-zero emission target in 2065 for its energy sector. In addition to increased renewable generation, the

national strategy focuses on the introduction of clean cooking, and efficiency and electrification interventions to reduce energy demand growth across the building, industry and transport sectors.

Generation: Uganda's renewable energy potential is provided in Table 6.

Table 6: Energy resource potential for Uganda [13]

Resource	Unit	Exploitable Reserve
Geothermal	MW	450 MW
Hydropower	GW	2 GW
Solar Irradiation	kWh/m2/day	5.1 kWh/m2
Biomass cogeneration	GW	1.65 GW
Peat	MW	800

The Uganda Electricity Generation Company Limited (UEGCL) is a parastatal organization that holds and operates the majority share of Uganda's installed generation capacity, contributing 913MW, which is equivalent to 68% of the market share. The breakdown of installed capacity per type of energy source for Uganda is provided in Figure 9 below.

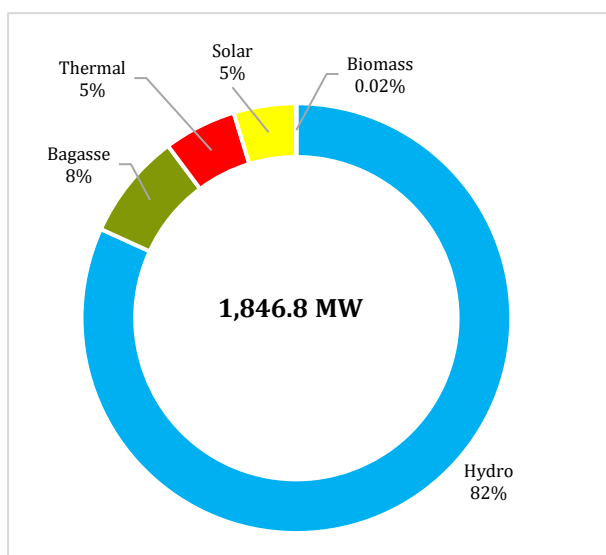


Figure 9: Uganda's Current Generation Mix [13]

By 2023, Uganda's generation capacity was 1,846.8MW [13], including on-grid and off-grid resources, and is likely not to reach the National Development Plan III goal of 3,500MW by 2024-2025 [14]. According to the Energy Transition Plan, Uganda expects to increase its installed generation capacity to over 40GW by 2050. This assumes an annual electricity generation capacity growth rate of 14%, while retaining the high share of renewables.

Demand/Consumption: Uganda's gross domestic electricity generation by grid-connected plants was 6,032 GWh in 2023, representing a 512 GWh increase compared to the previous year [15]. The transmission energy losses, calculated as a percentage of the energy purchased from generators but not sold, are 4.8% [16]. Despite the

high share of renewables in Uganda's electricity generation capacity, biofuels, waste, and oil account for over 97% of the final energy consumption in Uganda. Biomass is the primary source of energy; rural areas use wood fuels for household cooking, while urban areas use charcoal. The high rate of biomass use for energy supply has accelerated Uganda's deforestation, while fuel wood scarcity and increasing prices of charcoal have created additional challenges for rural areas. Additionally, biomass use in households is largely inefficient, with primitive appliances that have efficiency rates of ~10-20%

Transmission/Distribution: Uganda Electricity Transmission Company Limited (UETCL) owns, operates, develops, and maintains the high voltage transmission grid. It comprises 36 substations and has a total

length of approximately 4,518.66km. The distribution of electricity, particularly for the grid assets operating at 33kV and below, is handled by UEDCL. The total length of distribution lines is 63,986km, and comprises 76 distribution substations. The number of grid-connected customers is 2,083,444. Grid distribution losses have decreased substantially from 30% to 16% after the liberalization of the sector. To achieve universal electricity access in Uganda by 2030, 6.1 million new customers need to be connected to the grid, requiring at least USD 5.5 billion in investments. Public funding is also essential to address a USD 329 million affordability gap for Tier 1 solar home systems over ten years [17].

Electrification Strategy: Presently, 60% of Ugandans now have access to clean energy (45% for electricity and 15% for clean cooking) [18]. Access to electricity is concentrated in the capital city and the central region, which have developed grid infrastructure. However, around 6 million of those with access to electricity consume less than 50-75 kWh of energy per household per year (i.e., the International Energy Agency minimum for basic electricity service). The electrification strategy for Uganda focuses on several key programs to expand access to electricity. These include grid extension, grid densification, use of ready boards, and implementation of the Electricity Connection Policy (ECP). The latter commenced operations in November 2018 and aims to provide 300,000 new connections each year, free of charge for connections that either require no pole or 1 pole, with a goal to achieve a 60% electricity access rate by 2027. This electrification target has now been updated through Uganda's Energy Transition Plan, which has a 2030 universal access target where new connections will be established for 800,000 households each year. This annual electrification rate is similar to the rates achieved previously by other countries such as Kenya, Rwanda, and Bangladesh.

The Rural Electrification Agency has identified 683 mini grid sites that can each serve over 10,000 people. To further extend electricity access, the Government of Uganda is piloting 25 mini-grids in the northern, southern, and western regions of the country.

2.5. Review of previous studies relevant to the EAPP region

1. **2014 EAPP Masterplan (2015 – 2040)** [19]: The masterplan has a regional focus with a limited view into country-level analysis. Balmorel has been applied to developing the EAPP Master plan by analysing the benefits of regional cooperation and recommending a package of new cross-border transmission lines. The analysis includes 12 countries (i.e., Burundi, Djibouti, DRC, Egypt, Ethiopia, Kenya, Rwanda, Sudan, Tanzania, Uganda, South Sudan, and Libya), all considered in one model. The model covers a large area and a long-time horizon, from 2015 to 2040 in five-year time steps.

The master plan considers a main scenario to determine the least cost integrated generation and transmission plans for each country, as well as 20 variant scenarios. The main scenario assumes a target domestic generation capacity level for each country equaling 110% of its peak annual demand. Optimal generation and transmission are computed for each of the scenarios separately. The main differentiating factors considered in the variant scenarios and the sensitivity analysis were the following: limitations on transmission interconnectors, nuclear development, increased share of VRE in the region, CO2 emission limits, delays in hydro investments, exports to South(ern) Africa Power Pool (SAPP), lower costs for long-term hydro projects, as well as variations in fuel prices, interest rates, electricity demand, and availability of hydro inflow. Further analysis is required to compare the scenario results.

Similar master plans have been developed for the other African power pools. The common ‘main’ scenarios across the power pools are the benchmark scenario (based on existing and committed interconnections), the regional integration scenario, and the autonomous development scenario. This gives a detailed insight into the interactions and dynamics in the regional electricity system. Examples of parameters changing across scenarios are: Electricity demand, fuel prices, delay of projects, interest rate, cost of transmission and generation, CO₂ price, and targets for the share of renewable energy. The scenarios are used to analyse the robustness of different investments. This is important because of the uncertainty regarding the future development of many important parameters. In many studies, investment in generation and transmission is analysed separately, e.g., these compute the amount of transmission needed given a certain portfolio of generation. In the study, the model is investing simultaneously in generation and transmission. All scenarios find optimal solutions, and when a parameter is changed, the optimal investment in both generation and transmission will change. Only already defined investments in plants and transmission lines have been included as “existing” and “committed”. This gives more flexibility to the model to identify the least-cost investments. Predictably, this (together with the regional as opposed to national focus of the study) may result in different regional power system planning projections compared to the ones prescribed in the national Master plans of the individual countries.

2. **AfDB New Deal of Energy for Africa (2016 – 2030)** [20]: In 2019, the African Development Bank (AfDB) conducted an energy sector analysis to determine the energy investment needs for each country and to provide an overview of the future energy system at the power pool and continental levels. The analysis considered four scenarios: (i) reference scenario with minimal cost, optimized power supply for the regions, (ii) low carbon scenario introducing a carbon price, (iii) trade stagnation scenario without integrated power markets and new interconnector developments, (iv) BAU access expansion scenario where electricity access targets are reduced. The results projected an increased diversification of the EAPP’s generation mix, as well as an increased regional integration driven by generation surpluses and deficits in several areas of the region. The low-carbon scenario led to an overall 30% increase in the annual investment requirements for the forecasted period, with a considerable effect on the EAPP due to its heavy reliance on natural gas and coal generation. The trade stagnation scenario had a limited impact at the continental level, but the Eastern, Central, and Western Africa regions are more impacted due to their higher volumes of trade.
3. **IRENA Planning and Prospects for Renewable Power** [21] : The study’s analysis for the Eastern Africa region considered six scenarios covering low and high levels of VRE deployment, regional integration, and hydropower resource availability. The scenario results show peak carbon emissions around 2030, an overall reduction of system costs, and increased system flexibility with increased VRE deployment and regional trade.
4. **Promoting better economics, renewables, and CO₂ reduction through trade: A case study for EAPP** [22]: In 2020, a Remy et al. paper investigates the advantages of regional electricity trade in Africa through the use of a least-cost generation and transmission capacity model. Three integration scenarios are analyzed:
 - BAU Scenario: Countries independently optimize their generation expansion plans based on their specific demand, without additional coordination for dispatch and system planning beyond current practices.

- Shallow Integration: Each country develops its least-cost generation plan while optimizing trade along existing and committed interconnectors.
- Tight Integration: This involves a deep level of integration, thus requiring significant levels of coordination in expansion and operation.

The analysis indicated promising prospects for the EAPP with a "Shallow" level of integration, potentially achieving \$7.6 billion in benefits with respect to the BAU scenario, while retaining national plans and limiting interconnections to existing links. In the "Tight" integration scenario, where generation and new interconnection plans are optimized regionally, the benefits with respect to the BAU scenario could increase to \$18.6 billion. This approach is most efficient in meeting a 30% CO₂ emission reduction target by 2030, retaining \$11.8 billion of benefits. Importantly, this CO₂ target can be achieved at a cost 3.8 times lower (\$6.6 billion) than the \$25.7 billion required in the BAU scenario with very low trade.

Additional findings indicate that imposing a 20% upper limit on imports would still allow retaining 80% of the benefits, removing concerns about national energy security. The analysis also considers extreme drought conditions, which may reduce trade benefits due to 20%-30% lower hydro availability in hydro-dominated systems like Ethiopia. More than 60% of the base case benefits are retained under Shallow integration. Tight integration, with its higher flexibility, allows for a benefit retention of 89% or more by adjusting trade flows. This approach enhances the system's ability to mitigate major drought impacts and meet policy constraints, including CO₂ emission and import restrictions, at a lower cost due to a stronger and wider network.

5. **Resilience of the Eastern African Electricity Sector to Climate-Driven Changes in Hydropower Generation** [23]: In 2019, Sridharan et al. developed a framework that combines the use of long-term models for electricity supply and water systems management to assess the sensitivity of the benefits of potential expansion plans in the Eastern African Power Pool (EAPP) to the effects of climate change. They consider variations in water availability due to changing climate conditions and their impact on hydropower infrastructure operation.

The analysis revealed that the most resilient strategy for the EAPP corresponds to a plan optimized for a slightly wetter climate compared to historical trends. This finding suggests that infrastructure investments should take into account climate change projections, as planning for a wetter climate offers the lowest maximum regret value across different climate futures. By doing so, the EAPP can minimize the financial impacts of climate change on electricity generation costs.

The study further highlights the varying degrees to which different EAPP countries are affected by non-adaptation in the electricity sector. Countries like Egypt and Kenya, which rely on gas, geothermal, and coal for their electricity supply, experience minimal impact due to their low reliance on hydropower. On the other hand, Uganda and Tanzania could expect costs to vary significantly between the wettest and driest climate scenarios, with cost ranges of -18% to +28% and -5% to +23%, respectively, when compared to the baseline. The analysis underscores the importance of considering country-specific vulnerabilities and resilience in developing adaptation strategies and demonstrates the potential for substantial fluctuations in electricity costs under the no-adaptation scenarios, highlighting the vulnerability of the current electricity infrastructure to climate change impacts.



Overall, the study emphasizes the importance of climate-proofing infrastructure investments and developing resilient strategies in the expansion planning of the Eastern African electricity sector. It highlights the need to consider climate change uncertainties, country-specific vulnerabilities, and the optimization of adaptation strategies for a slightly wetter climate to minimize financial regrets and ensure a more resilient power system within the EAPP.

3. Modelling Focus Areas for EAPP Region

Case Studies

Below are the key characteristics, unique features, and focus areas relevant to the modelling and scenario development for the case studies planned for the EAPP region.

- **Electrification of Household Demand Segments** – With increasing urbanization, grid access, and economic development, the electrification of various demand segments is increasing across the region. Households represent the largest energy demand category across nearly all the EAPP countries, except Egypt, Libya, and South Sudan, where the transport sector consumes more energy. However, their electricity demand footprint has typically been much lower. Beyond challenges in grid access, household electricity demand in grid-connected areas remained low due to the prevalent use of other energy sources for energy needs. Therefore, increasing electricity demand from households will be an important factor considered in the demand projection for the energy system analyses in the case studies.
- **High biomass usage** – Traditional energy sources such as wood fuel and charcoal constitute the largest share of energy supply across several countries within the EAPP. Cooking and lighting are the primary loads at the residential level. However, in countries such as Uganda and Ethiopia, over 94% of household cooking demand is met through biomass-fuelled cooking. While Kenya has higher levels of liquefied petroleum gas (LPG) and kerosene use, biomass-fuelled cooking remains high (~67%). Moreover, the efficiency of the biomass-fuelled cooking appliances is often as low as 10-20%.
 - There are several initiatives underway within the EAPP countries to improve the efficiency of cooking appliances and to also introduce electric cooking. As such, projections for future demand should account for efficiencies in the demand segment as well as increased electric cooking loads from households.
 - Biomass is primarily used for household cooking, with households representing the largest demand segment in the region. It will be important to reflect household cooking demand as a major demand category in the energy system analysis case studies.

Consequently, household cooking, primarily through low-efficiency biomass cookstoves, is a key characteristic that will be represented in the base year for the country-level energy system analysis in Ethiopia, Uganda, and Kenya. Additionally, the transition from biomass to cleaner cooking fuels (i.e., electric cooking, LPG, and kerosene stoves) will be relevant to consider under the different scenarios for the energy system.

- **Low electricity demand per capita** – Electrification projects and experiences in rural areas show that rural households typically use electricity for limited purposes (i.e., lighting, radio, mobile charging). These loads are energized for a few hours, resulting in small electric energy use and making it economically unviable for utilities to do last-mile connections and extend the national grid to these areas. To improve the economic viability, there is a need to increase the number of productive uses of energy (PUE) appliances for cold storage, irrigation, and other agricultural machinery. However, some research also shows that household electricity demand will remain low across sub-Saharan African countries despite electrification efforts unless it is also accompanied by economic growth. As their Gross Domestic Product (GDP) increases, the elasticity of demand increases, and

demand increases more rapidly. As such, while households are currently characterized by low electricity demand, this is expected to increase as the GDP per capita grows (thereby increasing the affordability of electricity and electric appliances). This consideration will be relevant when considering energy demand projection data for the energy system analysis.

- **Integration of energy-efficient technologies** – The adoption of energy-efficient technologies, such as e-cooking appliances or efficient biomass stoves, can reduce greenhouse gas (GHG) emissions and household demand. Similarly, new efficiency measures and policies for the industry sector will reduce total demand. These changes will affect projections for future energy demand and supply, infrastructure needs, and environmental impacts. Different scenarios will be considered in the case studies to evaluate the impact of stronger energy efficiency policies on the national energy systems.
- **Demand growth** – Accurate demand projection is essential to validate the need for additional generation and transmission capacities. There have been significant variations in the demand projections for the EAPP and country-level analyses conducted in different studies. Relevant drivers for demand growth across the EAPP include population growth, urbanization, increasing electricity access, and increased demand due to commercial and industrial sector growth. In the recent EAPP masterplan, the electricity demand forecast is based on projections from each member country. It assumes an annual growth rate of 6% up to 2030, and 3% up to 2040 for the BAU scenario. Energy demand projection data used in the BAU scenarios for energy system analysis through GENeSYS-MOD should also be aligned with these growth trends.
- **Electrification of transport** – As part of the objective set in the Climate Resilient Green Economy Development (CRGE), East African countries like Ethiopia, Kenya, and Uganda are prioritizing green mobility to provide clean air and build healthier cities for their residents. Additionally, the high cost of imported gas and fuel scarcity concerns provide an enabling environment for electric vehicle (EV) adoption.
Since Ethiopia's recent ban on Internal Combustion Engine (ICE) vehicle importation for private use, the latest data from the Ministry of Water and Energy shows EV automobile ownership has reached over 8,800 in 2024. In neighbouring Kenya, tax incentives for EVs have similarly resulted in a spike in electric motorbike ownership, where the total number of electric 2-wheelers and 3-wheelers is estimated to equal 5,047 [24]. As governments continue to set ambitious EV adoption targets, tax incentives, and preferential electricity tariffs, similar trends are expected in Kenya, Uganda, Ethiopia, and other sub-Saharan countries. However, the increasing electrification of transport demand and the associated infrastructure needs for charging will present unique considerations and challenges for East Africa. In this regard, it is important to study the implications, challenges, and opportunities of increased EV adoption on generation, distribution, and transmission capacities.
- **Addition of New Energy-intensive Industries** – Several countries across the EAPP region have plans to introduce new energy-intensive industries, some of which are listed below. These new demand segments, which have typically not been accounted for in previous modelling studies on the region, and their impact on the energy and power system, should be considered in the case studies.
 - ICT sector (data centers, cryptomining, ...),
 - Manufacturing industries,
 - Special economic zones
 - Agriculture PUEs (mechanization, irrigation, ...)
 - Mining sector
 - Other PUEs.

- **High renewable integration** – Due to the high potential for renewable resources across the EAPP, most countries have set ambitious targets for increased generation from renewable sources. Hydropower and geothermal already represent the highest share of the installed generation capacity across Kenya, Ethiopia, and Uganda. Solar, wind, geothermal, and hydropower will be scaled significantly, while nuclear is also considered in some countries, such as Uganda. The committed generation expansion targets for renewables and the government commitments should be considered in the input data for GENeSYS-MOD under relevant scenarios.
- **Variability of Hydropower Resource** – As hydropower is one of the primary generation sources in the EAPP, variations in the hydro inflow will have significant effects on the energy balance and trade across the region. Hydro-dominated power systems are energy-constrained and depend on the hydrological conditions. Drought conditions should be accounted for when considering the effective capacity of hydropower plants to ensure the reliability of the system. Therefore, sensitivity analysis to examine the effect of drought conditions on the optimal generation expansion pathways will be very relevant to these countries.
- **Electrification Strategies** – Due to the low electricity access rates within the region (average of 45%), electrification is a major focus area for the region. The varied approaches considered for electrifying urban/semi-urban vs rural areas, as well as the mid-term and long-term strategies, should be accounted for in the generation expansion analysis. The main electrification strategies considered are listed next:
 - Grid expansion and densification are the primary approaches for urban/semi-urban areas.
 - Mini-grid and solar home systems (SHS) are considered as mid-term electrification strategies for rural areas (generation at the distribution level), accounting for 35-50% in some countries.
 - In the long term, the integration of mini-grids within the grid network should play a major role.
- **Losses of different types in the distribution and transmission network:** Poor performance of utilities with losses averaging 18-22% is typical within the countries in the region. Demand patterns and the use of resources such as biomass are largely affected by outages and interruptions in the electricity supply by utilities. This characteristic will be captured in the country-level energy system analyses: losses will be accounted for in the demand projection, along with expected improvements throughout the modelling period.

4. Case Study Specifications

4.1. Energy system analyses in Ethiopia and Addis Ababa

4.1.1. National energy system analysis for Ethiopia using GENeSYS-MOD

This will be the first case study analysis conducted by the Ethiopia team - a country-level energy system analysis for Ethiopia using GENeSYS-MOD. Two case studies will be carried out under this scope, with different modelling objectives and methodologies as discussed below. The first case study will be led by the Ethiopian Ministry of Water and Energy and Veritas. The second case study will be conducted by Addis Ababa University researchers. While both case studies will conduct country-level energy system analysis for Ethiopia to examine generation expansion pathways up to 2050, each case study will have different objectives, scenarios, and assumptions. General description of initial analyses planned.

Purpose: These case studies will analyse Ethiopia's national energy system to assess future pathways for 2050. The analyses will examine how different development trajectories, such as increased renewable integration, increased demand due to increased electrification of transport and household loads, climate resilience measures, and efficiency improvements, could impact Ethiopia's technology choice, generation expansion, economic costs, and GHG emissions.

Regional context: Ethiopia heavily relies on hydropower, making its energy system susceptible to climate variability. Moreover, Ethiopia has set a target to expand its generation capacity from the current 5.2 GW to 19.9 GW by 2030, through a diversified renewable technology portfolio. At the same time, electricity accounts for less than 3 percent of the total energy supply for the country; biomass is the primary energy source, especially for households. Given the projected increase in energy demand across Ethiopia, as well as the national and regional push for a diverse, more resilient energy system, the analysis will examine Ethiopia's pathways for generation expansion and technology choices up to 2050. These trends are similar across other EAPP countries – similar demand growth patterns are projected, while governments are pursuing targets for expansion and increased diversification of generation resources. Therefore, this analysis can also provide insights into scenarios that address both shared challenges and unique national goals in Ethiopia.

4.1.1.1. Main objectives and approach

Modelling Approach: Using GENeSYS-MOD, Ethiopia's energy system is modelled across multiple scenarios. Each scenario tests the implications of different policy choices and technological advancements, assessing outcomes in terms of energy mix, cost-efficiency, resilience, and sustainability. Sensitivity analyses explore uncertainties in water flow, capacity factors, costs, and demand changes, ensuring comprehensive interpretation and analysis of the future energy pathways for 2050.

Objectives:

- Analysis of Ethiopian government target scenarios vs BAU.
- Understand the implications of demand growth and electrification on the generation expansion plans for the country.
- To evaluate and compare Ethiopia's energy system performance under various scenarios, assessing how different policy choices affect investment decisions, energy security, cost, environmental impact (greenhouse gas emissions), and resilience.

- To provide actionable insights into Ethiopia's power pathway within the EAPP context, identifying strategies, policies, and best practices from other countries in the region.

4.1.1.2. Assumptions, scenarios, and input data considerations

Assumptions:

- **Future Cost Declines for Renewables and Storage:** Assumptions on the rate of cost reductions over time for solar, wind, and battery storage.
- **Long-Term Demand Growth Rates:** Assumptions about economic growth impacting electricity demand beyond currently available projections.
- **Hydrological Variability:** Assumed changes in seasonal water availability for hydropower due to climate impacts, particularly beyond historical records.
- **Policy Continuity:** Assumptions that current energy policies and incentives (e.g., subsidies for renewables) may or may not continue over the modelling period.
- **Regional data:** The nation-level analysis will use administrative regions to represent nodes within the energy system. However, it is understood that this may not precisely correspond to the actual electricity system within the country.

Scenario Descriptions: Various scenarios will be considered under the two nation-level case studies utilizing GENeSYS-MOD.

- **Baseline (BAU) scenario:** This scenario reflects a continuation of current policies, programs, and targets of the government on the energy sector. The baseline/BAU scenario will be the base of all other scenarios and assumes historical trends will continue in the future.
- **High Renewable Energy Integration Scenario:** This scenario will account for the national targets outlined in the utility's target for 2030, which are entirely composed of renewable energy generation technologies. The scenario will compare the renewable penetration rate and analyse the driving factors. In addition to the investment requirements, the Levelized Cost of Electricity (LCOE) and emission reduction potentials will also be compared.
- **High Electricity Demand Scenario:** This will account for new policy targets for EV and clean cooking adoption. Additionally, the implications of new demand categories such as irrigation and data centers will be analysed.
- **Efficiency scenario:** Efficiency improvement measures will be applied on both demand and supply sides, potentially through demand-side management, loss reduction, and grid enhancements.

Sensitivity Analyses

- **Climate resilience:** To understand the adaptability of each generation system to climate stressors.
 - **Drought impact scenarios:** Simulate conditions of severe drought to assess power system resilience and economic consequences. Extremely low inflow and thresholds where backup generation system/technology might be required will be evaluated.
 - **Seasonal water flow variability:** It involves analysing the impact of seasonal changes on hydropower generation, reflecting Ethiopia's reliance on water resources. Thus, varying the amount of the water inflow and differentiating the sensitivity of the model's outcome to the water inflow pattern will be studied.
- **Capacity factor sensitivity:** This will consider how variations in the capacity factor of hydropower plants (due to changes in inflow) impact overall generation. Reduced capacity factor scenario for the major hydropower plants will also be studied.
- **Sensitivity due to change in demand growth rate:** Varying energy demands per sector relative to the base case, and studying the model outcome.

- **Cost sensitivity:** Examines how fluctuations in technology and fuel costs impact overall system costs and investment viability.

Input data types

The models need extensive socio-techno-economic datasets, but it is also possible to work with limited data by following optimal approaches & analytical capabilities, as well as taking feasible assumptions. This is done considering the challenge of limited data availability in Ethiopia and other developing countries in general. Data in the case studies will be based on extensive data collection from various stakeholders, mainly from the Ministry of Water and Energy and Ethiopian Electric Power, but also relevant local and international reports will be used as data sources.

- **Demand Projection:** The two case studies will utilize different modelling tools (MAED and LEAP) to generate demand projection data for Ethiopia.
- **Historical Data:**
 - **Energy Demand:** Historical electricity consumption data, broken down by sector (residential, industrial, commercial, and transport).
 - **Resource Availability:** Data on Ethiopia's renewable resources, including hydro, solar, wind, and biomass potential.
 - **Existing Infrastructure:** Information on the current capacity of generation plants, transmission and distribution networks, and system losses.
- **Technical Data:**
 - **Existing energy system:** i) energy balances: overall aspects of the composition of the demands and an overall view of total energy supply, ii) power sector: data available from the utility on the individual plants and electricity transmission and distribution, installed capacity, load factors, efficiency, etc.
 - **Future energy system:** i) future technologies: known international sources, supplemented by nationally specific options where available. ii) future demands: projected based on existing demand projections or building relationships between drivers, trends, and disaggregated sector-based estimation of the national economy.
 - **Renewable Energy Technology Performance:** Actual capacity factors and operational efficiencies for solar, wind, and hydropower plants.
 - **Hydropower Generation Data:** Seasonal and long-term water flow data, including historical trends in river flow variability and their impact on hydropower generation.
 - **Transportation Data:** Vehicle ownership data will be used to estimate transport energy demand for both ICE and EV vehicles.
 - **Power Export:** The national utility has set export targets based on existing and ongoing interconnection projects; these will be considered for export projections.
 - **Energy Balance Data:** The national data reflecting total energy supply for all energy forms (power, fossil fuels, ...) will also be used.
- **Cost Data:**
 - **Capital and Operating Costs:** Real cost data for different generation technologies (e.g., solar, wind, hydro) as well as expected operational costs over time.
 - **Fuel Prices:** Current and projected prices for any fossil fuels that may be part of the system.
 - **Financing and Discount Rates:** Current market rates or assumptions based on recent studies for financing costs and discount rates used in capacity expansion modelling.
- **Economic Indicators:**

- o **Policy and Regulatory Data:** Existing national policies regarding renewable energy targets, electrification, energy efficiency programs, and emissions reduction goals, as well as the region's electricity trade agreements under the EAPP.

4.1.1.3. Expected outcomes

The following are the expected outcomes of the country-level energy system analysis.

- **Energy mix projection** forecasting the share and capacity-mixes of renewables (hydro, geothermal, and VRE) and other conventional sources till 2050, considering government policies and demand growth trends.
- **Resilience and adaptability analysis** assessing how the system will perform under various stress scenarios, particularly during droughts or periods of water flow variability, based on historical hydropower performance data. The diversification of generation resources and its impact on system resilience will be evaluated.
- **Cost-effectiveness analysis** detailing the system costs and emissions for each scenario, using current cost data and emissions factors for renewable and conventional generation technologies.
- **Identifying regional best practices** through analysis of the EAPP region countries and identifying best approaches and practices that can be translated to Ethiopia.
- **Policy and infrastructure recommendations** providing insights for future energy policy and infrastructure investments, based on comparative analysis across scenarios and sensitivities. This will inform the generation expansion targets and technology prioritization currently set by the national utility.

4.1.2. City-level Energy System Analysis for Addis Ababa

Two case studies will be conducted for the local-level energy system analysis for Addis Ababa. The first case study will be led by the Ethiopian Ministry of Water and Energy and Veritas. The second case study will be conducted by Addis Ababa University researchers. Each case study will have different objectives, scenarios, and assumptions. Input data from the utility will be commonly used across the case studies, where available.

4.1.2.1. General description of the initial analyses planned.

Ethiopia's ongoing energy transition from biomass and fossil fuels to electricity, driven by factors such as shifting demographics, rapid urbanization, modernization, and economic growth, presents a complex and evolving energy landscape that puts urban areas at the forefront of energy consumption and sustainability challenges. Similar to the nation-level analysis, the two modelling tools (MAED and LEAP) will be utilized to generate an annual energy demand projection for Addis Ababa up to 2050.

This research intends to conduct a city-level energy planning analysis for Ethiopia's capital and largest urban center, Addis Ababa, by soft-linking two open-source tools: Long-range Energy Alternatives (LEAP) and Global Energy System Model (GENESYSMOD). The energy-demand model developed in LEAP will be used to carry out analysis of the current and future energy landscape by diving-deep into the intricate tapestry of energy needs across various sectors of the economy – residential, commercial, industrial and transport sectors. The study seeks to contribute valuable insights to the discourse on the city-level energy system by providing low-carbon pathways for Addis Ababa's transition towards a more sustainable and resilient system. This research mainly focuses on the energy demand and electric distribution infrastructure of Addis Ababa.

This analysis will specifically focus on GHG emission reduction and efficiency improvement strategies by providing a quantitative assessment of the socio-economy, demography, technological change and future governmental directions. The outputs of the LEAP demand model will be an exogenous input to

GENeSYSMOD to investigate the capability of the substations and electric distribution system to meet the future demand. Different studies [25] [26] [2] show that there is poor technical performance in the distribution network of Addis Ababa, mainly due to old & traditional infrastructure, resulting in high distribution losses, frequent supply interruptions, and poor power quality. To cope with insufficient hours of service and power outages, households use backup solutions (biomass, fossil fuel-based solutions) for cooking & lighting, which in turn contribute to GHG emissions. This study considers policy interventions that should result in better service reliability, reduced losses, and good power quality in the near future.

The second case study will similarly utilize a soft-linked tool (MAED), which utilizes assumptions on socioeconomic, technological, and demographic development to generate energy demand projections used as input for GENE-SYS-MOD. The case study will analyse the urban development plans and policies currently considered for Addis Ababa city and their implications on the required generation expansion, costs, and emissions.

4.1.2.2. Main objectives and approach

The primary objective is to assess how Addis Ababa's energy landscape might evolve under various alternative policy scenarios. The approach includes disaggregated and sector-wise representation of the energy demand by customer groups, namely: residential, commercial, industry, and transport sector.

In the LEAP-GENeSYSMOD case study, the demand representation for the residential sector follows a bottom-up approach, by using different appliance classifications (lighting, cooking, baking, refrigeration, other loads). Such an approach is believed to enhance the accuracy of the energy demand representation, analysis, and policy implications. In addition, building scenario-based models to explore alternative pathways and quantify the potential impact of policy targets on energy consumption, efficiency, and greenhouse gas emissions will help policymakers prepare for unexpected developments.

The MAED-GENeSYSMOD case study will focus on understanding the implications of policy and development targets set forth by the current city administration. These will be composed of targets for e-mobility, clean cooking, energy efficiency, and grid service improvements.

GENeSYSMOD will be used to simulate different scenarios, evaluating the effects of government policies focused on energy efficiency, service reliability, supply adequacy, and infrastructure capability. By comparing these alternative policy scenarios, the analysis aims to reveal the extent to which the city's energy system growth, service quality, and environmental goals can be achieved.

4.1.2.3. Assumptions, scenarios, and input data considerations

The data used for the study results from extensive data collection from various stakeholders, mainly from government institutions, and published data sources in the literature. Specific data sources include the Ministry of Water and Energy, the Ethiopian Electric Utility, the Ministry of Transport, and the Ethiopian Statistics Agency. Key assumptions include rates of growth for various factors such as socioeconomics, GDP, per-capita income, and population, which are derived from regional and national development plans, as well as international projections.

In this study, we employ different scenarios for the projection of the energy demand and also explore the capability of the distribution network to supply it. The scenarios are mainly drawn with the assumption that future policies that promote efficiency improvement, emission reduction, and supply reliability will be implemented. The reliable supply scenario considers targeted interventions to minimize the duration and frequency of interruptions.

Scenarios considered:

- BAU scenario: This scenario will utilize historical data and provide a baseline for the generation mix, capacity growth, technology choices, and investments for Addis Ababa's energy system. This scenario will be considered for both of the case studies.



- City policy target scenario: Addis Ababa is part of the C40 network² and has made notable commitments for its urban energy system; these targets, along with the city's broader development plans, will be considered under this scenario. This scenario will be considered for the case study utilizing the MAED-GENeSYSMOD workflow.
- High demand growth scenario: Different policy, socio-economic, and technological factors contributing to high demand growth will be considered under both case studies. This will, for example, consider low biomass utilization in urban households as a result of policy and economic shifts.
- Energy efficiency scenarios: Moderate and ambitious target scenarios will consider moderate and aggressive interventions to maximize energy efficiency and emission reductions in both the demand and distribution levels. This scenario will be considered for the case study utilizing the LEAP-GENeSYSMOD workflow.

4.1.2.4. Expected outcomes

The city-level energy analysis is expected to generate quantitative estimates of the economic and social benefits of transitioning to a more reliable and sustainable energy system. Decision-makers will gain access to policy recommendations that balance infrastructure investments with the anticipated benefits of reduced outages and improved service. These findings will help guide long-term energy planning for Addis Ababa and serve as a model for similar urban centers across the country and beyond.

The analysis is expected to show how Addis Ababa's energy profile could change under different government target scenarios. Key outcomes include projections of energy demand, potential reductions in greenhouse gas emissions, and the efficiency gains achievable through targeted policy interventions. These findings will help determine the feasibility of government goals and offer actionable insights for policymakers regarding the investment priorities, urban development strategies, policy development, and the potential impact of transitioning towards a more sustainable urban energy system.

4.1.3. Rural Electrification in Ethiopia Using OnSSET

4.1.3.1. General description of initial analyses planned

Ethiopia's rural electrification efforts are an essential element of the country's broader energy transition, aimed at reducing reliance on traditional biomass and fossil fuels while enhancing access to modern energy services. Despite progress in expanding national electricity infrastructure, over 55% of the population—predominantly in rural areas—still lacks access to electricity [27]. Addressing this significant gap requires innovative, data-driven strategies tailored to Ethiopia's unique geographical, demographic, and socio-economic conditions. This study investigates the least-cost pathways to achieve universal electricity access using the Open-Source Spatial Electrification Tool (OnSSET) till 2030. It enables the optimization of the yearly deployment and use of electrification technologies by incorporating a variety of data inputs, including energy demand, resource availability, population density, terrain, and proximity to existing grid infrastructure [28], [29]. This approach evaluates a combination of grid extensions, mini-grids, and standalone systems (e.g., solar home systems) to determine the most viable solutions for each settlement [30], [31], [32]. The analysis will explore how these solutions can address Ethiopia's unique rural electrification challenges, such as dispersed settlements, low population density, and limited existing

² Addis Ababa city is a member of the C40 Cities – a network of mayors from leading cities globally, that are working to address climate change. Addis Ababa has made several targets as part of its draft Climate Action Plan and its C40 engagement; these include targets for CO₂ emission reduction, air quality improvement, and climate resilience.

infrastructure. It will also assess the financial investments required and identify zones that should be prioritized for electrification, particularly in areas where interventions can deliver the highest socio-economic benefits. Moreover, this study recognizes the importance of aligning electrification planning with Ethiopia's NEP 2.0 [2] targets, particularly grid densification and intensification for automatic grid connection for settlements that are in proximity to the grid.

4.1.3.2. Main objectives and approach

The main objective of this research is to explore optimal pathways for rural electrification in Ethiopia under different policy scenarios. The study focuses on addressing two critical research questions: (1) Where are the optimal geographical locations to expand grid infrastructure or deploy off-grid systems? (2) What are the additional connections, generation capacity, and investment requirements needed to achieve universal electrification? The answer to these questions will provide essential input to effective policy and investment decision-making; the analysis will inform the geographical scope of access improvements across different communities and regions of the country. To achieve these objectives, the study employs a structured multi-phase methodology. The first phase involves extensive data collection from open-source resources to compile geospatial, demographic, demand, and techno-economic datasets. Nighttime satellite imagery is utilized to map unelectrified rural settlements, providing a spatial representation of electricity access gaps. These data are given as input to the OnSSET model to simulate least-cost electrification pathways, enabling a granular analysis of the technologies best suited for each settlement.

Electricity demand estimation plays a central role in the methodology. The Multi-Tier Framework (MTF) is employed to model variations in electricity needs across rural and urban settings, accounting for differences in consumption levels and service quality expectations [7]. Demand projections are further refined using key demographic indicators such as population growth rates, household sizes, and GDP growth. Techno-economic considerations, including grid connection costs and capital expenditures for off-grid technologies, are incorporated to assess the financial viability of various solutions. This comprehensive approach ensures that the study provides actionable insights for policymakers aiming to expand energy access cost-effectively and sustainably.

4.1.3.3. Assumptions, scenarios, and input data considerations

The study will incorporate a range of input data, including geospatial data, demand, demographic, and techno-economic data. Key inputs will encompass population density, land cover, topography, proximity to existing infrastructure, and resource potential (e.g., grid infrastructure, mini and small hydropower resources, solar irradiance, wind speeds). These factors are essential for determining viable sites for renewable energy solutions. Population growth rates and urbanization are assumed to follow the UN population projection. Techno-economic data, such as capital and operational expenditures for grid connections and off-grid systems, are included to calculate the LCOE, facilitating comparisons of the cost-effectiveness across technologies. Electrification is a dynamic and complex process that varies over time. The results of Pappis et al. [33] suggest that a technology change could occur in the future in many locations in Ethiopia. Thus, this study will evaluate three scenarios to explore the implications of different possible universal access targets and different policy pathways on rural electrification. In this regard, the Current Trend Scenario (CTS) assumes that electrification continues at the current growth rate. This scenario represents a baseline, reflecting a BAU approach with no significant policy shifts or accelerated efforts. In contrast, the Sustainable Development Goal Scenario (SDGS) aligns with the United Nations Sustainable Development Goal 7 (SDG7), which aims for universal access to modern, affordable, and reliable energy by 2030. The SDGS assume a substantial acceleration of electrification efforts, supported by strong policy interventions and financial resources. Finally, the Grid Densification Scenario (GDS) builds on Ethiopia's NEP 2.0 ambition, emphasizing grid expansion and densification [3].

4.1.3.4. Expected Outcomes

The study is anticipated to illustrate what Ethiopia's rural electrification looks like under different scenarios, such as the CTS, Sustainable Development Goal Scenario (SDGS), and Grid Densification Scenario (GDS). Under each scenario, it will identify the optimal mix of electrification technologies, including grid extensions, mini-grids, and standalone systems, for each rural settlement. The study will also provide the estimation of the number of new connections required to achieve universal electrification, along with a detailed breakdown of the additional generation capacity required to meet the future electricity demand and investments needed to achieve universal access. The analysis will also prioritize electrification zones, enabling policymakers to allocate resources effectively in scenarios of budget constraints. By addressing Ethiopia's unique rural electrification challenges, this research offers a scalable framework that can inform energy access strategies across similar contexts in Sub-Saharan Africa.

City-level electric mobility planning in Addis Ababa

Two case studies will be conducted for the city-level electric mobility in Addis Ababa: i) private EV mobility planning and ii) electric bus mobility planning; these are discussed below. Beyond these, a third case study utilizing the EV-PV tool for a comparative analysis of optimal charging infrastructure deployment across Addis Ababa, Kampala, and Nairobi is under consideration.

4.1.4. Private electric vehicle mobility planning in Addis Ababa

4.1.4.1. Background and general description

Ethiopia, as a country, has been experiencing the effects of climate change in the form of flooding, erratic rainfall patterns, drought, average temperature, heatwaves, and wildfires. Knowing this, the Government of Ethiopia, as part of the CRGE Development, has set as one of its objectives the development of the transport sector green by promoting fuel-efficient and environmentally friendly transportation [34]. Besides the emission, another reason to discourage the use of petroleum-based transportation and move to the electrification of road transport is the significant increase in the price of fuel oil. Ethiopia relies on imported Petroleum products, and the global spike in the price of oil has inflated the import bill, which makes it very challenging, given the recent forex shortage in the country.

The electricity price in Ethiopia is also one of the lowest in Africa, with an average rate of 4.5 US cents/kWh [35] owing to the predominant and cheap hydropower generation source, added with government subsidy to low-income communities. With abundant renewable resources (hydroelectric, wind, solar, and geothermal), Ethiopia currently has an installed capacity of 5,200MW, with the potential to generate over 60,000 Megawatts of electric power.

With this background, Addis Ababa - the capital of Ethiopia - emerges as a key city for studying the electrification of privately-owned vehicles. Not only does it stand out for its political and economic importance, but also for symbolizing a dual narrative, with significant opportunities as well as critical challenges. On the one hand, Ethiopia's strong commitment to EVs suggests that a rapid transition to electric mobility in Addis Ababa is highly plausible.

4.1.4.2. Objectives, approaches, scenarios, and outcomes

This study follows a first general methodological framework tailored to the local context and conducts an initial high-level analysis for Addis Ababa. The goal is to develop and demonstrate the relevance of a model tailored to African cities by identifying crucial factors needed to effectively address electric mobility planning problems within the local context. It also aims to quantify the spatio-temporal charging demand

under diverse charging scenarios. This will help uncover key challenges and give preliminary recommendations for effective private EV deployment in Addis Ababa.

Leveraging geospatial data, our approach allows us to estimate the daily charging needs along with their spatio-temporal dependence. Subsequently, the model allows for an analysis of the implications on local energy infrastructures (such as charging station deployment and load profiles) along with the assessment of local photovoltaic potential to meet demand. Specifically, the Electric Vehicles-photovoltaics (EV-PV) model will be used to assess the above issues by answering the following three research questions:

- What is the current and future additional electric demand required by the charging of electrified private vehicles, and how is it likely to be distributed in space and time?
- What are the implications for Addis Ababa in terms of public charging infrastructure requirements and potential impacts on the local power system (i.e., number and location of stations needed to meet demand effectively)?
- How can locally installed photovoltaics help to meet the additional demand and mitigate potential drawbacks on the power grid?

The EV-PV model will analyse three different archetype charging scenarios: i) predominantly charging at home, ii) predominantly charging at workplaces, and iii) a mixed scenario involving charging at home, workplace, and points of interest. For each, the spatio-temporal charging needs will be analysed in detail (aggregated on traffic zones and on a city-wide scale), along with the implications on the required charging stations and the potential for PV energy to meet the charging needs.

4.1.5. Electric Bus Mobility Planning in Addis Ababa

4.1.5.1. General description of initial analyses planned.

This study intends to optimize the deployment of smart charging stations for electric buses in Addis Ababa with the goal of improving customer satisfaction and utility advantages. The study begins by evaluating Addis Ababa's present electric bus charging infrastructure, taking into account variables including the number of charging stations currently in place, traffic patterns, and the capacity of the electrical grid. Accessibility, demand trends, and closeness to renewable energy sources are all taken into consideration when identifying possible sites for new charging stations using Geographic Information Systems (GIS) and optimization models. In order to achieve sustainability and lessen the impact of EVs on the current power system network, the research also takes into account the integration of renewable energy source generation into the charging station.

Creating a framework for the thoughtful placement of charging stations that strikes a compromise between the demands of utility companies and EV customers is one of the main goals, as is coming up with regulations to facilitate the installation of these stations. The study utilizes open-source Python and GIS-based electric mobility planning modelling tools: i) the EV-PV (Electric Vehicles-Photovoltaics) and ii) the Electric Vehicle Fleet Simulation Model (EV-fleet-SIM), which correlates the required number of charging stations with the necessary capacity for charging EVs. Furthermore, the research adopts a mixed methods approach, incorporating quantitative data from existing electric bus users alongside qualitative insights from stakeholders in the transportation and energy sectors.

4.1.5.2. Main objectives and approach

The main goal of this study is to compute the optimal deployment of smart electric bus charging stations in Addis Ababa, by improving customer satisfaction and utility advantages.

- To assess Addis Ababa's current electric bus usage and charging habits, as well as the infrastructure readiness for smart charging stations.

- To develop a framework for optimal deployment of smart charging stations, particularly for electric buses, considering future penetration, utility, and user benefits till 2030.
- To propose strategies for integrating renewable energy sources into the existing charging stations in the city.
- To evaluate the economic, technical, and environmental impacts of the proposed deployment strategies.
- To provide conclusions and recommendations for the best location of electric bus (EB) charging stations.

This research begins by reviewing academic sources on EV infrastructure, smart charging technologies, and urban mobility to assess the current state of EV adoption and charging infrastructure in Addis Ababa, identifying knowledge gaps and challenges specific to the city. And then, critical data, including traffic patterns, demographics, the electric bus market, grid capacity, and renewable energy potential, will be collected and analysed. Using this data, an optimization model based on EV-PV and EV-Fleet-SIM will be developed to identify the most effective deployment strategies for smart charging stations. Finally, the outputs of the model will be evaluated to assess the economic, technical, and environmental impacts of these policies, culminating in actionable recommendations to establish an efficient and sustainable charging station deployment policy for electric buses in Addis Ababa.

4.1.5.3. Assumptions, scenarios, and input data considerations

The study will use various input data, including geospatial, EV, and infrastructure data, to optimize smart charging station placement. This will cover charging station locations, road networks, traffic, bus routes, and renewable energy potential.

Geospatial Data:

Geospatial data is vital for planning smart charging stations, covering the locations of existing and potential stations near bus routes, terminals, and depots. It includes road network maps with traffic patterns, congestion hotspots, and bus route geo locations.

Electric Bus Data

- Number of electric buses in Addis Ababa
- Number of other EVs in Addis Ababa (optional for broader studies & used to estimate adoption rate)
- Energy consumption, distance travelled, and time required to fully charge.
- Daily operating hours of electric buses
- Average number of charging sessions per day per bus

Public EV Charging Infrastructure

- Number of currently available public EV charging stations
- Capacity of each charging station (kW per charger)
- Types of chargers available (AC, DC fast chargers, etc.)

City and Traffic Data

- Traffic congestion levels by area and time (e.g., peak hours)
- Road network details (length, road types, intersections, etc.)
- Bus route data (frequency, routes, and stops)
- Average speed of buses during operation

Renewable Energy Resources

- Solar energy potential data in Addis Ababa (e.g., average daily sunlight hours)
- Solar power generation potential per square meter

- Current or planned solar panel installations at charging stations.

By organizing this data, we will guarantee a solid dataset that facilitates efficient planning and optimization of the deployment of smart charging stations.

This research will consider multiple scenarios to ensure the solutions for optimal deployment of smart charging stations are robust and adaptable. Below are the key scenarios that can be explored:

1. **Baseline Scenario: (current infrastructure and demand):** Analyzes the present state of electric bus infrastructure in Addis Ababa, including bus routes, energy demand, and grid capacity. This establishes a foundation to compare improvements and explore gaps in current systems.
2. **Future Growth Scenario:** Models the impact of future increases in electric bus fleets as part of government policies from 2020 to 2030. This assesses the scalability of the proposed charging infrastructure under projected demands.
3. **Transition to Fully Electric Bus Fleet Scenario:** This scenario will explore the impact of transitioning all current public buses in the city to electric buses (EB). This will assess the infrastructure, energy demand, and environmental effects of such a shift.
4. **Renewable Integration Scenario:** Evaluates the feasibility and impact of deploying renewable energy generation (e.g., solar) in the charging stations using EV-PV and the System Advisor Model (SAM), one of the dependency software of EV-Fleet-Sim. This explores sustainable options to reduce reliance on the grid and lower operational costs.

By considering these scenarios, the research will be comprehensive, addressing immediate needs and long-term strategies for an effective and sustainable smart charging network tailored to electric buses in Addis Ababa.

4.1.5.4. Expected outcomes

The expected outcomes of this research could include:

- **Optimal Deployment Plan:** A strategic planning input for effective policy and pathways to help guide policy makers and energy planners for deploying electric buses in Addis Ababa.
- **Energy Efficiency Analysis:** Insights into energy consumption patterns of electric buses and strategies to minimize energy use.
- **Infrastructure Requirements:** General estimation of the required number of charging stations, charging capacities, and integration with renewable energy sources.
- **Impact Assessment:** Evaluation of the impact of electric buses on urban mobility, energy demand, economic and environmental sustainability.
- **Policy Implications:** Suggestions for policies to support the adoption of electric buses and renewable energy integration in public transport systems, and recommendations for scaling up electric bus adoption and charging infrastructure based on current and future demand.

4.2. Replication energy system studies in Uganda

4.2.1. General description of the initial analyses planned

Purpose: The analysis of Uganda's energy system will assess the set of government targets. Uganda's energy system will be studied for the 2050 planning horizon, with a base year of 2018 and time steps considered for 2025, 2030, 2035, 2040, and 2045. The different case studies will utilize GENeSYS-MOD, openTEPES, Plan4res, and EV-PV for the analyses. This will include the assessment of scenarios such as those focused on increased renewable energy integration, the decarbonization of the energy system, the electrification of the railway system, the deployment of the required storage capacity for intermittent source generation, and the minimization of the economic costs. The geographic scope of the case studies is further detailed below.

Country context: According to the Uganda Energy Transition Plan, electricity demand is expected to reach around 200 TWh by 2050. The demand is currently met by clean source generation, with 90% served by hydropower generation. Uganda's plans to diversify its generation mix include geothermal, nuclear, wind, and solar [36].

4.2.2. Main objectives and approach

Main Objectives

- To assess the Uganda Government's set power and energy targets.
- To assess the performance of Uganda's energy system under multiple scenarios, evaluate the required installed capacity and cost.
- To identify strategies and policies necessary to achieve government-set targets for Uganda's Energy system.
- To assess the interconnection of the Uganda and Kenya power systems

Modelling Approach: Uganda's energy system will be modelled using GENeSYS-MOD, assuming different scenarios. These include: the set of yearly emission targets, an increased renewable energy integration one, one on the electrification of the railway system, and one focused on the reduction of transmission losses. Uganda is split into four regions - the Central region, the Western region, the Northern region, and the Eastern region.

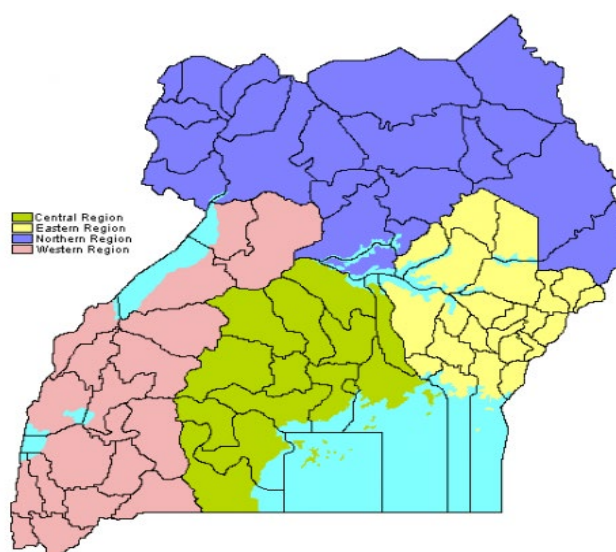


Figure 10: Regions of Uganda

Modelling focus: The broad modelling focus areas and the details of Uganda's case studies under the OpenMod4Africa project are described below.

A. Generation analysis

A decarbonization approach for the energy system in Uganda: According to Uganda Energy Transition Plan, the access rate to clean cooking still remains at 15% with only 5% of users relying on liquified petroleum gas or electricity, largely in urban areas, and the other 10% relying on improved biomass cookstoves. Additionally, the energy sector emissions are projected to increase in the next decade. However, the government has committed to reducing its economy-wide emissions by 5.9% compared to its baseline scenario in 2030 in the updated Nationally Determined Contribution [36]. The assessment will involve the comparison of the projected energy demand from cooking, transportation, industry, and the installed capacity from clean energy sources.

Generation security assessment under extreme climate conditions: With 95% of Uganda's installed energy capacity coming from renewable sources such as solar, biomass, combined heat and power, and hydropower, these energy sources are inherently intermittent. They are also more sensitive to extreme climate conditions. For instance, severe droughts in Northeastern Uganda could reduce water levels, limiting hydropower generation in the region. Similarly, heavy rainfall during certain periods of the year could impact solar power generation in the Central, Eastern, and Western regions. Given these challenges, it is essential to analyse generation security in extreme climate scenarios. Thus, this analysis will focus on ensuring energy security in the affected regions through strategies like the interregional energy transfer or increasing local energy reserves. Such measures would be vital as Uganda aims to increase its installed capacity to 70.5 GW, with intermittent renewable energy sources contributing 91%.

B. Transmission network assessment

Analysis of the impact of the new electricity act policies/power sector reforms on the development of the transmission network in Uganda: Uganda's electricity transmission infrastructure faces significant limitations, which restrict the integration of new power generation projects and hinder the expansion of energy access. With Uganda's growing renewable energy capacity and the aim to establish sustainable solutions like e-mobility and smart grids, a robust transmission network is essential for ensuring that generation projects can connect seamlessly to the grid and supply power reliably. This analysis will examine how Uganda's energy policy influences the development of its transmission network, focusing on key areas such as grid expansion, renewable energy integration, and resilience to climate challenges. It will also explore opportunities for aligning the network's growth with Uganda's long-term energy ambitions, such as achieving a 45 GW solar capacity by 2050 and supporting emerging sectors like EV charging infrastructure.

Assessment of the long-term integration of renewable technology generation in Uganda's transmission network: Uganda's demand is currently met by clean sources, with 90% from hydropower generation. With a projected demand of about 200 TWh in 2050, Uganda plans to diversify its generation to include geothermal, nuclear, wind, and solar [36]. The required transmission and storage capacity for the intermittent source generation, including solar and wind, will be assessed on a scenario basis.

C. Uganda-Kenya power systems interconnection study

An assessment of the interconnection of Uganda's and Kenya's power systems: This assessment will study the impact of this interconnection on the generation targets. The study will also determine the optimal generation dispatch for the interconnected power systems. The analysis will also examine relevant outcomes that can inform the region-level planning at the EAPP.

D. Electrification of road and railway transport systems

Insights into an optimal EV charging infrastructure deployment model for Uganda: With the Government of Uganda's recent investment in e-mobility, including the establishment of Kiira Motors Corporation, an EV manufacturer, it is essential to assess the capacity of the current transmission and distribution networks to handle the increased loads resulting from the installation of DC fast chargers. These chargers draw significant power from the network within a short period, which could strain existing infrastructure.

To address this, the study will leverage the EV-PV tool, starting with a local case study on Kampala. This case study will assess the EV charging demand, determine the required number of charging stations, and evaluate the potential of PV-based charging solutions in mitigating grid stress and meeting part of the charging needs. By identifying optimal locations within the network that can support the installation of these chargers while ensuring grid stability and reliability, this model can serve as a blueprint for replication in other locations across Uganda.

A feasibility assessment for electrification of the Uganda Railway network: According to Vision 2040, Uganda will have a railway system with high-speed trains for both passengers and cargo by 2040. This will link Uganda to four routes to the sea through Mombasa, Dar-es-salaam, Djibouti, and Tanga ports [37]. A feasibility analysis of the electrification of the projected railway network will be carried out to assess the demand and required installed capacity.

4.2.3. Assumptions, scenarios, and input data considerations

Assumptions

- Energy demand growth: Assumptions on the growth of industrial, domestic, and transport with clean energy sources.
- Change in transmission losses.
- Increased integration of renewable energy sources

Scenario descriptions

- **Baseline scenario:** The current generation mix, investments, and total emissions of the energy system.
- **Renewable energy integration scenario:** assessment of the utilization of renewable energy resources to meet the transport, industry, and domestic demand, and the resulting emissions reduction.
- **Electrification of the railway network.**
- **Increased capacity of storage and transmission options** for intermittent sources such as solar and wind.
- **Efficiency improvement scenario:** Efficiency improvement, including the reduction of transmission losses.

Input data considerations:

- **Historical Data:**
 - **Energy Demand:** Historical power system demand data and demand growth projections.
 - **Resource Availability:** Renewable resources availability, including hydro, solar, wind, and biomass potential.
 - **Generation data:** Time series generation data from the current installed capacity, including hydro, solar, thermal, and combined heat and power.

- **Technical Data:**
 - **Power system data:** Existing generation capacity from various sources, transmission networks, and transmission losses
 - **Technology Performance:** Availability factors, capacity factors of solar, wind, and hydropower plants.
 - **Efficiency improvement:** Sensitivity analysis will be considered for efficiency improvements, such as the reduction of transmission losses.
- **Cost Data:**
 - **Fuel costs:** current and projected costs of fossil fuels used in the energy system.
 - **Capital costs:** cost of installing and constructing different generation technologies and transmission infrastructure.
 - **Operating cost:** Cost of operating the different generation technologies and transmission infrastructure.

Data sources

- Uganda Energy Transition Plan [36]
- Petroleum Authority of Uganda
- Electricity Regulatory Authority
- Uganda Electricity Transmission Plan (2018-2040)
- Uganda Electricity Generation Company
- Uganda Electricity Transmission Company
- Uganda Electricity Distribution Company

4.2.4. Expected outcomes

The expected outcomes from the analysis include policy recommendations, an energy mix projection, an assessment of infrastructure requirements to meet specific targets, and the definition of suitable pathways to achieve the targeted emission reductions.

4.3. Replication energy system studies in Kenya

4.3.1. General description of initial analyses planned

Description of planned analysis

Purpose: The Kenya energy system studies aim to consolidate the country's energy development strategy to drive inclusive and sustainable economic growth. This initiative is anchored in Kenya's developmental priorities, Nationally Determined Contributions (NDCs), and the Integrated National Energy Plan (INEP) framework. The INEP framework provides a strategic roadmap for ensuring energy access, sustainability, equity, and alignment with Kenya's long-term socio-economic goals.

The plan focuses on stimulating economic activity across all counties by reducing dependence on the export of key commodities such as ammonia, jet fuels, and gasoline. Instead, it seeks to harness the country's vast renewable energy resources, including its green hydrogen potential, to transition various sectors—particularly transportation (road and railway)—to clean electricity.

Key elements of this transition analysis include projecting future energy demands, evaluating existing resources, and assessing the capacity to meet increased demand. The plan also prioritizes economic feasibility, compliance with NDCs, adherence to international agreements, and integration with regional energy markets. Furthermore, it incorporates a just transition framework to ensure fairness by addressing the needs and concerns of local energy entrepreneurs and stakeholders. By leveraging the INEP framework, Kenya aims to establish a resilient and sustainable energy system that supports its economic transformation while addressing global climate goals.

Regional context: Kenya boasts significant geothermal reserves, offering a unique opportunity to position the country as a regional hub to produce green ammonia and green hydrogen, particularly for the aviation and marine sectors. Efforts to electrify Kenya's railway system, which is closely interconnected with Uganda's network, underscore the importance of regional coordination in advancing sustainable transportation and trade between the two nations.

As a member of the EAPP, Kenya prioritizes regional energy trading to drive economic growth and energy integration. Leveraging wheeling services, the country seeks to optimize cross-border energy flows, attracting investors and enhancing energy market efficiency. These initiatives will serve as a foundation for developing and analyzing regional energy scenarios, supporting a resilient, interconnected, and sustainable energy future for East Africa.

4.3.2. Main objectives and approach

Modelling Approach: GENeSYS-MOD modelling can be used to explore different scenarios that are guided by the development plans, NDCs, and the country's approaches through a development framework, the available resources, and the regional market strategy. These scenario analyses should focus on adequately representing the priority sectors to assess the investment requirements and the development approaches.

Objectives: The objectives and the approaches to model Kenya's energy scenarios are;

- a) To assess a comprehensive BAU scenario using existing energy models, considering current energy demand patterns and network infrastructure data, while ensuring alignment with regional energy planning requirements.
- b) To assess the country's energy needs and development plans through the engagement with key sector stakeholders and a review of existing reports, providing a basis for defining national targets and the activities required for developing precise modelling scenarios.
- c) To develop and evaluate these modelling scenarios with a focus on integrating strategic plans, addressing the national priorities, and aligning with regional and international commitments. We

should refine these scenarios through stakeholder engagement and the dissemination of model results.

- d) To develop guidelines and a framework for model updates.

4.3.3. Assumptions, scenarios, and input data considerations

The following modelling scenarios will be considered for Kenya's Case Studies.

Transition to Electric Rail in the Transportation Sector: Kenya's Vision 2030 includes a strategic plan for railway transport outlined in the 2023-2027 roadmap, aiming to deliver world-class rail services through the development and modernization of a safe, reliable, and sustainable rail network. Currently, the railway system serves approximately six million passengers annually, but it fully relies on fossil fuels, which are not only environmentally harmful but also contribute to higher transportation costs due to low efficiency. Additionally, because the fuel is entirely imported, Kenya's energy security remains at risk. To realize this vision, the country is moving towards transitioning to electric trains, which will help reduce emissions, lower operational costs, and increase energy independence. This scenario assesses the projected electricity demand growth, generation response, and the comprehensive expansion of the power system needed to support this transition.

The **assumption** made is that the country will be able to start shifting to electric railway by 2027, first focusing on 50% of the system by starting this transition with Meter Gauge Railway and then including Standard Gauge Railway (SGR) by 2030. The power generation capacity will have to increase by boosting local generation to supply the added load.

Input data considerations: the most relevant input data include the number of annual passengers, which will increase according to the population growth rate, as well as with the increase in the attractiveness of this service, the energy consumption per railway passenger per kilometer, and the costs: i.e., the capital, fixed (electricity tariff), and variable ones.

Advancing Renewable Energy for Sustainable Economic Growth in Communities: To achieve Kenya's Vision 2030, the Ministry of Energy, in collaboration with development agencies, has established the INEP to guide sustainable energy development across the country. Under Kenya's devolved governance structure, each of the 47 counties is now responsible for creating its own energy plan, which will be integrated into the national framework. These county-level plans are essential for promoting economic development by leveraging locally available renewable energy resources.

Scenario studies within this framework will be used to assess the community-level economic activities, evaluate the energy demand, identify the local renewable resources, and analyse the current grid connectivity. These studies aim to outline pathways for energy development within each county and determine the necessary grid extensions to support the distribution of energy among counties.

The **assumptions** made include the fact that the community activities will run on off-grid technologies and that the growth rate will be determined by the population and GDP growth rates.

Input data considerations: the main input data include the set of identified community activities, the energy demand for them, which will increase with the GDP, and the costs and features of the several technologies to use and deploy, including storage units. This is in accordance with the country plan to prioritize the county energy development plan under the INEP framework.

Expanding the Grid and Transitioning to Clean Energy with Integrated Storage Solutions: Kenya is dedicated to reducing its reliance on polluting fuels by transitioning the electricity generation to that based on clean energy sources and enhancing power reliability. A key focus of scenario modelling will be evaluating the demand for energy storage solutions and their impact on expanding energy access. This

assessment will compare various storage technologies, examining the implications of their deployment and use for energy investments and infrastructure development. The findings from these analyses will inform proposals for a sustainable energy market, supported by targeted policy recommendations to drive effective implementation.

The main assumptions made in this scenario analysis include the following: i) deploying storage, among other things, to provide the inertia, given the lack of enough resources providing this in the system when significantly increasing the intermittent clean energy sources in the network.

The input data specifically considered for this scenario follows: i) limits imposed on the use of fossil fuels and other GHG emission resources; ii) parameters for storage solutions deployed.

Transition to Sustainable Local Production of Green Fertilizers: To reduce reliance on foreign imports and foster economic growth through industrialization and agricultural modernization, Kenya is prioritizing the development of green hydrogen projects. It is essential to investigate the impact of this transition on the use of energy and water resources, as well as on energy investments. This assessment will aim to define economically, socially, and environmentally sustainable approaches to ensure a just transition that supports Kenya's development goals.

The **assumptions** made that are specific to these analyses include: i) the use of green hydrogen produced using electricity from the grid and off-grid solar PV generation to meet the local demand for green ammonia. In the base year, we will consider high utilization of the current PV generation capacity for this activity. The capacity devoted to this will grow incrementally using the GDP.

The input data specifically considered for this scenario follows: i) the green hydrogen specified annual demand.

Green Hydrogen for Aviation and Marine Transportation Services: This scenario analysis will determine the green hydrogen demand for aviation and marine transportation services, the corresponding required power generation capacity, and the CO₂ reductions achievable through the clean transition of these services. Within these analyses, the structures and institutions that will be necessary to support the successful realization of this vision will be identified and characterised.

The main **assumptions** made when defining this scenario include the following: i) there is a gradual transition to green hydrogen in aviation and marine transportation starting in 2030, whose characterization is provided within the input data section.

Main specific **input data:** i) amount of green hydrogen to be produced for these transport uses and the associated parameters; ii) the aviation and marine transport schedule.

Transition to Solar Photovoltaic (PV) Charging for EVs: Kenya's Vision 2030 considers the general transition to e-mobility. This scenario will assess the energy demand for e-mobility and evaluate the potential of solar PV to meet these needs. A comprehensive analysis of the requirements for this shift, along with the exploration of various development models, will be essential to support the successful implementation of the transition of the transport sector.

4.3.4. Expected outcomes

The expected outcomes for these case studies include the following: i) a set of tools whose use has been finely tuned to conduct these highly relevant analyses; ii) a set of relevant scenarios on the transition of Kenya's national energy system according to the development plans defined; iii) a set of policy proposals supporting this transition; and iv) a set of investment plans comprising the investments required to make the various transition pathways possible.

4.4. Power system analyses in the EAPP region

4.4.1. General description of initial analyses planned

Power system analyses will be conducted for the EAPP region to determine the optimal expansion of the power systems in the region, considering the need for sustainable, efficient, and cost-effective planning to meet the region's growing electricity demand.

The analyses will cover 13 countries, which are members of the EAPP, over a 10-year time horizon.

4.4.2. Main objectives and approach

The main objective of the study is to analyse a cost-efficient expansion of the power generation and the related transmission capacity in the region.

OpenTEPES will be used for the power system analyses in the EAPP region. The OpenTEPES model is a decision support system for defining the integrated generation, storage, and transmission expansion plan of a large-scale electric system for a 10-20-year horizon. The model will be used for the study, and four scenarios will be investigated. These scenarios are explained in the next section.

4.4.3. Assumptions, scenarios, and input data considerations

The input data for this task will be collected from different sources. Most of the data will be provided by EAPP. However, some of the inputs for Ethiopia, Kenya, and Uganda will come from the country-level analyses conducted within the project. For any other data types for which data is not available from the above two sources, generic parameters will be used.

Four different scenarios will be covered in the power system analysis for the EAPP region.

1. **BAU:** In this scenario, it is assumed that the future power demand will evolve according to historical trends. This will be closely aligned with the demand forecasts of EAPP's different member countries, as well as the Power Pool master plans. Furthermore, the candidate planned and committed to generating assets and transmission lines that will be drawn from the Power Pool's master plan. This scenario will be used as a benchmark for the other scenarios in this study. Additionally, the BAU will be modelled in SPLAT-AFRICA and will be used to validate and fine-tune the OpenTEPES model.
2. **Increased demand:** This scenario considers larger demand growth than in the BAU scenario. The key demand growth drivers in this scenario include universal access to energy, the higher energy intensity of activities in households and businesses, the higher economic growth rate, and an increased EV penetration rate.
3. **Reduction in reservoir levels due to climate change impact:** In this scenario, the impact of climate change on the inflows and water levels in reservoirs will be assessed. As detailed in Section 2, the hydropower generation in the EAPP region is significant, particularly in DRC, Ethiopia, and Uganda.
4. **Increased penetration of variable renewables:** In this scenario, the impact of increasing the penetration of variable renewable energy resources, such as wind and solar, within the EAPP region, will be investigated.

4.4.4. Expected outcomes

The output of the different scenario analyses will include the following: i) investment costs for the expansion of generation and transmission; ii) the marginal costs for the different generation technologies; iii) the total operational, emission and reliability costs; iv) the optimal operation of the system, including the unit commitment, startup, and shutdown decisions for non-renewable units, and the aggregated generation output per technology (thermal, hydro, RES, etc.); v) the amount of RES energy curtailment; vi)



the amount of transmission grid losses; vii) for hydropower plants, the evolution of reservoir volumes across the target year(-s), others.

4.4.5. Potential synergies with other models of the case studies

This task will have synergies with energy system analysis and rural electrification studies for Ethiopia, Kenya and Uganda, where models GENeSYS-MOD and OnSSET are to be used to analyse the national energy system transition. The EAPP power system analysis will use results from these analyses as inputs to conduct the relevant studies for the EAPP region.

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